

Graph Neural Networks for building and civil infrastructure operation and maintenance enhancement

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ABSTRACT

This systematic review, conducted within the PRISMA framework, investigates the disruptive capabilities of Graph Neural Networks (GNNs) in optimising Operations and Maintenance (OM) practices within the building and civil infrastructure domain. Addressing 5 research questions and encompassing 111 studies from 2014 to 2024, our study identifies the multifaceted applications of GNNs across different project stages from data enhancement to operational scenario enhancement. When considering integrated Facilities Management (FM) approaches, GNNs are employed for data enhancement purposes, leveraging techniques such as semantic enrichment of Building Information Modelling (BIM), various data imputation scenarios, and semantic segmentation of point clouds to enhance data quality and completeness. Operational scenarios involve the utilisation of GNN algorithms for anomaly detection, fault classification, system optimisation, and forecasting. Methodological optimisations crucial for GNN feasibility include feature engineering, architecture optimisation to balance complexity and overfitting risk, and the integration of Explainable Artificial Intelligence (XAI) methods to enhance model validity and trust. Physical principles integration through Physics-Informed Graph Neural Networks (PIGNNs) further enhances model explainability and validation. Future research directions focus on data interoperability enhancement, scalability improvements, and explainability enhancements. Automated graph generation and labelling, heterogeneous GNN models, supporting algorithms such as Long Short-Term Memory (LSTM) and reinforcement learning are proposed to overcome analysis limitations. Specific workflows targeting building performance-based semantic enrichment, building systems data imputation, and interdependency prediction are proposed in future directions. The review highlights the symbiotic relationship between GNN-based analysis and digital twin data analysis, emphasising the suitability of GNNs in addressing the demands of digital twin data analysis in the building and civil infrastructure domain.

1. Introduction

The recent rise in Artificial Intelligence (AI) has attracted widespread attention, fuelled by breakthroughs in Large Language Models (LLMs) and generative AI-driven products that offer simplified access to the public. Alongside these developments, Graph Neural Networks (GNNs) have emerged as a formidable Deep Learning (DL) algorithm gaining traction across various domains. Initially introduced in 2005 [1], GNNs were developed to overcome the limitations associated with topological data representation in traditional Machine Learning (ML). Traditional ML methods necessitate extensive pre-processing of topological data, which in turn carries the risk of losing crucial topological insights, ultimately leading to increased costs [1,2]. On the contrary, GNNs have

advanced DL as an algorithm specifically tailored to operate efficiently within topological or non-Euclidean spaces such as graphs [3]. In this context, graphs are structured data entities composed of nodes and edges, distinct from the conventional notion of 'graphs' as visual representations or plots. The versatility of GNNs is reflected in diverse domains such as navigational services and image recommendation platforms. Google Maps [4] utilise GNNs in predicting estimated times of arrival leveraging the intricate connections between various geographic points. Similarly, Pinterest [5] use GNN algorithms to deliver personalised image recommendations, tapping into the underlying graph structure of user interactions. GNNs have further gathered significant interest beyond consumer-facing applications in fields such as medicine and quantum chemistry in detecting adverse drug interactions by

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analysing complex medication networks [6], and in predicting unknown molecular properties [7] respectively. The expansion of GNNs represents a significant milestone in the evolution of AI, facilitating enhanced data representation and analysis across a spectrum of real-world scenarios.

Facilities Management (FM) constitutes an industry devoted to the strategic coordination of all physical assets within a built environment, aiming to optimise their performance, functionality, and sustainability. This encompasses the management of space, infrastructure, services, and personnel to reinforce the core business objectives of an organisation [8]. Operation and Maintenance (OM), a pivotal sub-domain of FM and the primary focus of the present study includes the day-to-day activities involved in guaranteeing the efficient operation and durability of these physical assets. This encompasses tasks such as monitoring, repairing, and upgrading, as well as optimising and forecasting asset-related conditions to ensure the optimal functionality and longevity of the built environment [9]. The integration of AI within the built environment aimed at enhancing the aforementioned workflows is becoming increasingly prevalent. The potential of such AI systems in managing building services and controls has been duly recognised [10,11], prompting further investigation into sophisticated AI techniques in areas such as energy efficiency management and fault detection and diagnosis, alongside the integration of technologies such as Building Information Modelling (BIM) and the Internet of Things (IoT) [12,13]. BIM serves as a foundational digital representation tool for physical systems [9], inherently interoperable with OM workflows. Integrating BIM with IoT, facilitates bidirectional communication between physical assets and their digital counterparts [14,15]. Civil infrastructure maintenance systems, likewise, present a wealth of heterogeneous and multi-modal data, including geometric, geospatial, and performance attributes [16]. When considering analysis, ML algorithms in particular have found widespread application in predictive maintenance [17], fault detection and diagnosis [18], and anomaly detection [19]. However, the incorporation of diverse data types, followed by the combination of multiple technologies, further adds to the complexity of the analysis of OM scenarios [20]. Unlike ML methods, DL-based methods such as GNNs are capable of handling challenges posed by multivariate, low-quality, and high-volume data [21,22]. As future research in integrating AI into OM systems hinges on the effective analysis of these data, attention must also be paid to the non-Euclidean nature of most of these data types [23–25] which GNNs are uniquely suited to address. Despite the proven versatility of GNNs across various domains, there remains a notable research gap in utilising GNNs building and civil infrastructure OM, prompting our exploration into this underexplored potential.

When considering existing reviews, the overall capabilities of GNNs, particularly the development of advanced algorithms, have been discussed in computer science-based review studies [2,26,27]. A range of studies [28,29] including a comprehensive review [25] have been undertaken within the construction domain to delineate the capabilities of GNNs for construction applications. Construction projects involve the creation of new structures in accordance with specific designs, adhering to building codes, and typically possessing well-defined commencement and conclusion dates. Conversely, OM concentrates on the maintenance and optimisation of existing assets, ensuring continual compliance with safety and environmental regulations, and necessitating ongoing efforts to ensure functionality and efficiency. From a task-oriented perspective, construction encompasses surveying, earthworks, and structural activities overseen by architects, engineers, and construction managers. On the other hand, OM involves inspections, maintenance, and upgrades managed by technicians and facility managers. Construction endeavours demand initial investments and confront unique challenges such as design alterations and delays. In contrast, OM initiatives involve ongoing operational and maintenance expenditures, further compounded by equipment failures and unanticipated maintenance occurrences. GNN-based analyses in construction harness BIM, project schedules, cost data, safety records, and resource data to optimise project deadlines, budgets, safety protocols, and resource allocation.

Conversely, OM prominently leverages real-time sensor data, maintenance logs, and asset performance metrics, prioritising longevity, uptime, and regulatory compliance [30,31]. Thus, the applications of GNNs in OM are distinctive and necessitate innovative explorations in feature engineering, considering the functional disparities compared with other industries.

In examining existing applications in data preprocessing or preoperational stage of building and civil infrastructure OM systems, GNN functionalities have been leveraged to enhance data quality through the addition of extra layers of information [32,33], and to augment data quantity by imputing missing values [34,35]. During the operational phase, other GNN-based functionalities such as anomaly localisation have been employed in system monitoring to identify potential anomalies [36,37], as well as in strengthening performance optimisation and forecasting future scenarios [37]. While GNNs for standalone OM tasks have been explored, documenting these techniques across both preoperational and operational project phases is notably absent. The added complexity of the OM domain further necessitates interdisciplinary research, bridging engineering, computer science, and data analytics [38]. Engineers from diverse disciplines are indispensable, particularly when tasks extend beyond the purview of facility managers. Simultaneously, computer scientists develop and maintain bespoke software solutions specifically tailored to meet the evolving needs of OM operations. In parallel, data analytics plays a crucial role by leveraging extensive datasets to derive actionable insights, thereby informing decision-making processes within the industry [39]. However, interdisciplinary studies that explore these domains concerning building and civil infrastructure OM are sparse [40,41]. We further aim to address this overall scenario, by identifying the distinct OM applications across both pre-operational and operational stages, identifying unique interdisciplinary developments and further pinpointing areas that have been overlooked or inadequately addressed.

The utilisation of GNN algorithms within a framework explicitly incorporating engineering knowledge to facilitate a comprehensive understanding of the generated outputs is essential [42,43]. Methodological advancements such as explainable AI (XAI) [44] and physics-based systems [45] could further validate the implementation of these DL algorithms within engineering systems. In further enhancing the validity of the outputs for engineering applications, conceptualising logical graph representations of physical systems is vital. The prevalent approach for such graph development is known as a structural scenario, entailing the depiction of physical layout and interrelationships within buildings or infrastructure in graph form. The innate graph-representable nature of these structures renders GNNs a natural fit for modelling and comprehending these environments [2]. This emphasis is highlighted by various studies, including those focusing on building spatial classification [32], inter-building energy optimisation [46], and the prediction of deterioration in water and road infrastructure [47,48]. However, the exploration of GNNs in non-structural scenarios remains relatively limited, presenting ambiguous graph representations due to the absence of clear physical layouts and relationships. This encompasses the investigation into scenarios such as adapting GNNs to model complex, dynamic relationships between images obtained from these systems [49], and correlated variables [50,51]. Furthermore, there is a notable absence of a foundational resource for guiding GNN system architecture development in the current domain [52]. Therefore, we aim to identify GNN architecture development with a focus on explainability enhancement tailored for OM to establish best practices for the implementation of GNNs in this domain. Thus, the underexplored application of GNNs, as algorithms capable of comprehending the complex OM data, the distinct characteristics of OM scenarios relevant to GNN application across preoperational, operational and interdisciplinary scenarios and the persistent requirement for GNN architecture development with a focus on explainability enhancement for engineering applications, collectively highlight the compelling need for conducting this review. We further consolidate these research gaps into the following research

question to be addressed in this study:

1. How can GNNs facilitate integrated OM scenarios from initial project phases?
2. What OM functions utilise GNN algorithms in the operations stage?
3. What feature engineering, architecture and hyperparameter optimisation make GNNs viable for the OM domain?
4. What advancements are being made in accommodating GNNs for engineering applications based on explainability enhancement?
5. What are the future research directions to accommodate GNNs for further OM enhancement scenarios?

1.1. Graph neural networks

A graph can be defined as a structured data representation that consists of nodes and edges. The formal definition of a graph G is a pair (V, E) , where:

- V is a set of nodes
- E is a set of edges, where each edge e is represented as a pair (u, v) , indicating a relationship from node u to node v .

A GNN is engineered to process data structured as a graph. While there exist various formulations and architectures, a prevalent foundational model is the neighbourhood aggregation or message-passing framework, as illustrated in Fig. 1. Throughout the subsequent sections of this study, the term ‘GNN’ will be utilised to denote various architecture in general, unless specific distinction is warranted. Mathematically, a GNN can be delineated in terms of its update and aggregation functions. Given a graph $G=(V, E)$ with node features X , the GNN operates in layers (or iterations). At each layer, every node aggregates information from its neighbours and possibly itself. Subsequently, the aggregated information is utilised to update the node’s features [2].

Aggregation function:

$$a_v^{(k)} = \text{AGGREGATE}^{(k)}(\{h_u^{(k-1)} | u \in \text{Neighbors}(v)\})$$

where:

- $h_u^{(k-1)}$ is the feature of node u from the previous iteration (or initial node features for $k = 1$).
- $\text{AGGREGATE}^{(k)}$ is a function that aggregates the features.

Update function:

$$h_v^{(k)} = \text{UPDATE}^{(k)}(h_v^{(k-1)}, a_v^{(k)})$$

where:

- $h_v^{(k-1)}$ is the feature of node v from the previous iteration.
- $a_v^{(k)}$ is the aggregated feature from the first step.
- $\text{UPDATE}^{(k)}$ is a function that updates the node’s features.

Aggregation involves gathering information from neighbouring nodes, aiming to capture local structures and features. Common aggregation functions include sum, mean, or weighted sums of neighbouring nodes’ features, providing a summary of a node’s immediate surroundings (up to 2 hops). Following aggregation, updating a node’s features typically involves employing neural network layers to combine aggregated information with the node’s current features, thereby generating a new feature set. This updated feature set aims to encapsulate both the node’s own information and its neighbours’ features. Message passing encompasses both aggregation and update steps, wherein nodes transmit messages (aggregated information) to neighbours for feature updates. Iterating this process facilitates information propagation across the graph, enabling nodes to capture information from distant parts of the graph. This mechanism integrates aggregation and update functions to facilitate efficient information exchange and propagation [53].

The primary functionalities of GNNs encompass node classification, which involves categorising entities within a network; graph classification, important in identifying the properties of entire graphs; and link prediction, which forecasts potential connections between nodes. GNNs also facilitate graph generation, thereby enabling the creation of new, structurally analogous graphs, and graph clustering, which identifies groupings within intricate networks. Furthermore, GNNs support node and graph regression tasks, wherein they predict continuous values for individual nodes or entire graphs, respectively. GNNs demonstrate proficiency in learning graph embeddings, effectively capturing essential structural information in a condensed format, and can leverage graph attention mechanisms to emphasise critical components pertinent to specific tasks. Additionally, they exhibit proficiency in handling dynamic graphs, enabling the prediction of future interactions or structural alterations, thus highlighting their extensive applicability within various domains [2].

1.2. Graph types in the building and civil infrastructure domain

Operating and maintaining buildings and civil infrastructure involves managing a complex web of components, systems, and interactions [54]. Graph representations of the relevant data can encapsulate these complexities by employing entities that reflect their physical or functional attributes [55]. While the data could be granularly categorised, simplifying them into three primary graph types of spatial, temporal, and interaction offers a concise yet comprehensive framework for capturing the essence of relationships within this domain. By logically combining these three types of graphs, more sophisticated representations such as spatial-temporal-interaction graphs can be

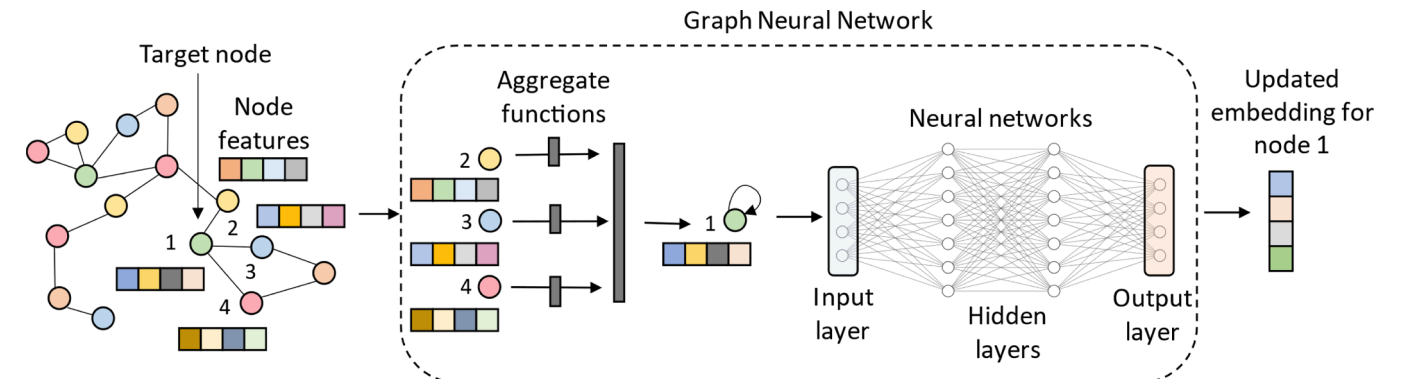


Fig. 1. Functional architecture of a GNN.

constructed [56]. These graph structures are then integrated into GNN-based analyses, typically through adjacency matrices [57]. At the core of these representations lies the connection matrix, derived from a spatial graph, which denotes the physical links between nodes. For instance, in a building's floor plan, rooms are interconnected by corridors, forming a network where rooms represent nodes and corridors represent edges [32]. GNN-driven spatial optimisation studies are developed by leveraging the features of these data points. This categorisation extends to geometric graph applications, such as 3D models of buildings or point cloud representations [58].

Temporal graphs are instrumental in capturing the dynamics of data or relationships between elements as they evolve over time [49]. For instance, building energy management systems can be represented in temporal graphs [59]. Nodes within these temporal graphs often symbolise energy consumption measurements taken at regular intervals, such as hourly readings. Meanwhile, edges illustrate the progression of these measurements over time. By utilising temporal graphs, facility managers can discern patterns crucial for energy-related forecasting. In the domain of infrastructure health monitoring, particularly concerning sensor-equipped bridge systems, temporal graphs offer a means to detect structural anomalies [60]. In this context, sensor readings serve as nodes, interconnected by their temporal sequence through edges. Leveraging subsequent analysis powered by GNNs, anomalies can be identified and addressed. Exploring deeper into this temporal dimension enables engineers to anticipate wear and tear over prolonged periods [60].

In further exploring graph data representation, interaction graphs concentrate on delineating the dependencies and interactions among disparate components or systems. These dependencies may not solely be spatial or temporal. For instance, within a building's HVAC (Heating, Ventilation, and Air Conditioning) system, an interaction graph would interpret how various variables impacting cooling load interrelate [50]. These graphs can be further extrapolated to encompass intricate systems such as multiple-building complexes. Here, nodes symbolise energy sources and consumers, while edges illustrate the transfer or dependency of energy between them. An integrated building management system harnesses spatial graphs (depicting physical layout), temporal graphs (illustrating system performance over time), and interaction graphs (highlighting system interdependencies) to enhance operational efficiency and streamline maintenance processes [51].

Upon defining the graph structure, the feature engineering process assumes paramount importance in defining node features, edge features, and global graph features. This phase is critical for effectively capturing the intricate interactions and dependencies within the system. The selection of features holds pivotal significance in the training of GNNs [61]. Factors such as material properties, geometric characteristics, or usage attributes guide the selection of these features based on their relevance to the given task. To ensure consistency and comparability, transformations such as normalisation or scaling of data are frequently applied to standardise features onto a similar scale. Moreover, categorical attributes such as material types are typically transformed into numerical formats using techniques such as one-hot encoding,

facilitating their integration into the feature set [32]. Subsequently, the resulting graph datasets, alongside their engineered features, are represented in matrix form, poised for utilisation in GNN-based training, testing, and validation procedures. This process often involves the utilisation of specialised DL libraries such as PyTorch Geometric, tailored specifically for handling graph data structures and operations [62].

1.3. The development of GNNs

The development of GNNs to advanced spatial-temporal models has been marked by the integration of sophisticated ML and DL techniques, complemented by strategic GNN architecture and hyperparameter optimisation. The development timeline is represented in Fig. 2 highlighting a few notable contributions from various domains, predominantly from computer science.

The foundational architecture of GNNs centres on capturing the global structure of graphs through iterative updates of node features based on neighbouring information [1]. Graph Convolution Networks (GCNs) build upon this foundation by introducing structured, convolution-like operations, leveraging the spatial/spectral structure of graphs to balance neighbour influence. Graph Attention Networks (GATs) were developed by incorporating dynamic attention mechanisms, permitting varying degrees of importance to be assigned to each neighbour's features, thereby adapting to the specific context of each connection [63]. GraphSAGE on the other hand was developed by introducing a neighbourhood sampling strategy that enables the model to learn from a fixed-size subset of a node's neighbours rather than the entire neighbourhood, thus reducing the computational complexity associated with each node and rendering the model scalable to large graphs [64]. Furthermore, Heterogeneous GNNs were specifically devised to process graphs comprising multiple types of nodes and edges, thereby facilitating learning from the intricate relationships and attributes inherent in diverse real-world data systems [65].

Dynamic GNNs expand the scope of GNNs to include temporal aspects, where relationships and attributes evolve over time [66]. Incorporating elements from variants of Recurrent Neural Networks (RNNs), such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) or Temporal Convolution Networks (TCN) [66], these models capture the evolution of graphs, considering the changing nature of both nodes and edges. Spatial-temporal GNNs and their counterparts, spatial-temporal attention GNNs, are combinations of the above-mentioned basic models. They are designed to incorporate data exhibiting spatial dependencies, attention mechanisms, and temporal evolution. These models often employ strategies such as pooling and sampling for computational efficiency and integrate with other ML and DL algorithms, forming hybrid models [67].

Equipped with a comprehensive grasp of foundational concepts, theoretical frameworks, fundamental graph types pertinent to the domain in question, and evolutionary developments, the following sections of this review are systematically organised to answer the research question outlined previously. In Section 2, we present the methodology adopted for sourcing and curating the relevant literature.

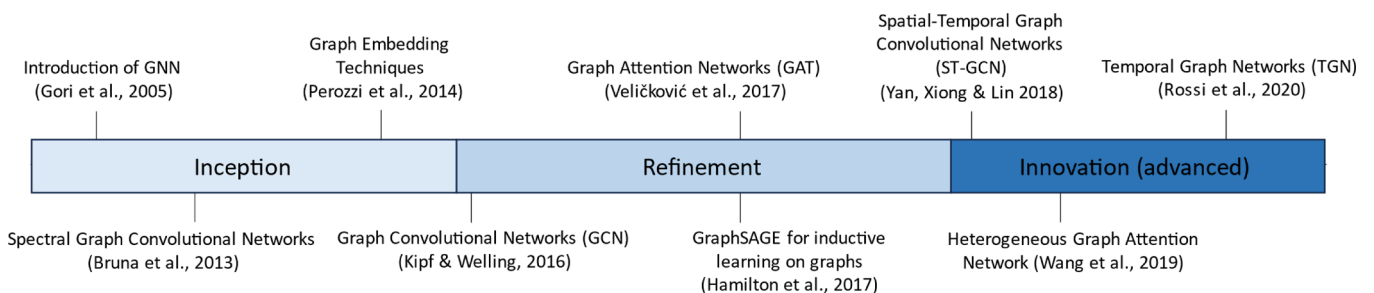


Fig. 2. The development of GNN architecture.

Following this, [Section 3](#) offers a comprehensive analysis of the current applications of GNNs. [Section 4](#) shifts the focus to discussing overarching research trajectories, pinpointing existing challenges, and exploring prospects in the field. [Section 5](#) brings the review to a close, summarising the findings and highlighting their significance in the broader context of GNN research.

2. Methodology

The current study aims to conduct a systematic review following the PRISMA framework [68]. Following established guidelines, the literature search was conducted across multiple databases [69,70]. Specifically, the Scopus and Web of Science databases were selected to ensure a comprehensive retrieval of relevant literature. Detailed search strings for both databases are provided in [Appendix A](#). The initial segment of the keyword string focused on different variants of GNNs. It was observed that GNN implementations are widely prevalent, with 58,038 publications spanning various disciplines such as computer science, engineering, and mathematics, as depicted in [Fig. 3](#).

The initial search was narrowed down to specifically target the subject areas of Engineering and Energy, which were the only relevant categories as defined by the Scopus database. A similar filtering mechanism was also followed for the subdomains in Web of Science that are closely relevant to building and civil infrastructure as shown in [Appendix A](#). The study includes both research publications and conference proceedings, recognising the rapid increase of GNN-based studies as demonstrated by [Fig. 4](#). This trend further highlights the growing significance and applicability of GNNs within the domain of engineering. The second part of the keyword criteria was used to narrow down the search to the building and civil infrastructure OM domain. Terms encompassing operation and maintenance were incorporated with various construction project phases, notably planning and design stages to identify studies that focussed on integrated Facility Management (FM) approaches during the project's preliminary stages. Literature that did not align with the definitions of "Operation" and "Maintenance" were systematically excluded at later stages. This expansive approach to keyword selection was adopted to prevent the undue limitation of the literature pool [71,72].

[Fig. 5](#) illustrates the systematic article selection process undertaken

for the literature review. Initially, a total of 2,949 studies were identified from both the Scopus and Web of Science databases. Following the removal of 316 duplicates using RStudio, the titles of 2,633 studies were screened. From this screening, 2,117 studies were excluded due to irrelevance to the research domain or because they addressed different models or abbreviations under similar names. Subsequently, abstracts of the remaining 516 studies were assessed, leading to the exclusion of an additional 405 studies for similar reasons overlooked during the initial title screening. The remaining 111 studies underwent in-depth evaluation and were deemed suitable for inclusion in the review. These studies encompassed 65 technology-focused studies, 31 application-focused studies, and an additional 15 studies centred on methodological advancements and theoretical frameworks. The technology-focused articles (65 in total) provide a comprehensive exploration of technological advancements, covering aspects such as GNN architecture, algorithms, training techniques, and evaluation methods. The 31 application-focused articles offer insights into GNN applications within the OM industry, showcasing practical use cases and real-world implementations. Additionally, the remaining 15 articles play a pivotal role in establishing the theoretical foundation for this study. A further classification was performed, delineating studies into two phases: preoperational or integrated FM approach-based studies and operational scenario-based studies. A thematic categorisation into data enhancement, system monitoring, performance optimisation and forecasting was also conducted to facilitate the analysis of application types.

The methodology of this systematic review, while rigorously implemented, presents certain inherent limitations. The utilisation of specific databases may have resulted in the involuntary exclusion of pertinent publications from non-indexed sources. Furthermore, the linguistic constraint to English-language publications potentially omitted relevant research conducted in other languages. Despite the implementation of an expansive keyword strategy, the possibility exists that some studies employing alternative terminology were not captured. Given the rapid pace of advancements in GNN research, there is a possibility that some recent developments were not fully represented in the literature at the time of this review. Additionally, the categorisation of studies into distinct focus areas (technology, application, methodological) inherently involves a degree of subjectivity. These methodological constraints should be taken into consideration when interpreting and generalising

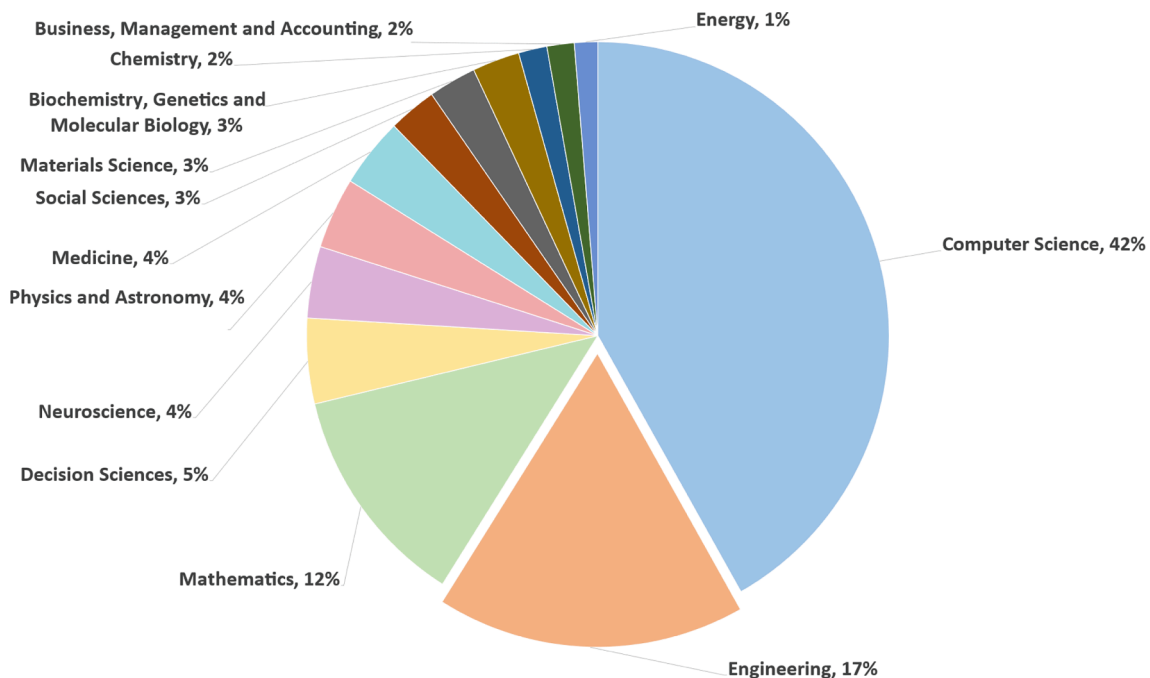


Fig. 3. GNN-related publications by subject area on Scopus.

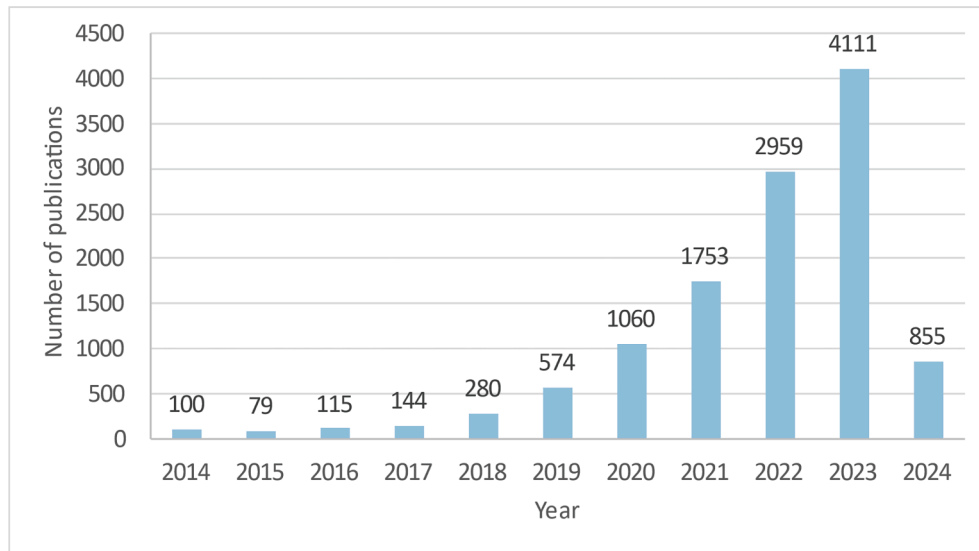


Fig. 4. Number of journal publications on GNNs in the engineering sector between 2014 and 2024 on Scopus.

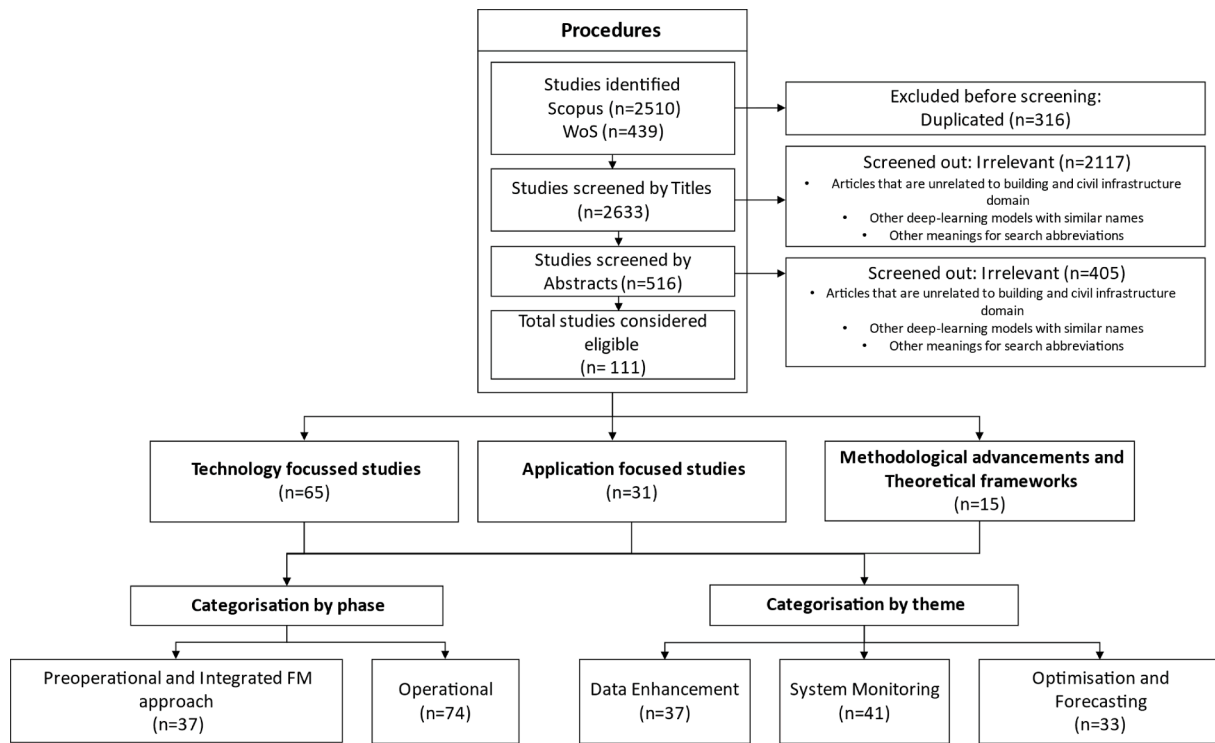


Fig. 5. PRISMA process.

the findings of this review.

3. Review of GNN applications in building and civil infrastructure operation and maintenance

The studies incorporated into this review were segmented into three principal categories: data enhancement, system monitoring, and performance optimisation and forecasting. The integration of AI in OM scenarios heavily relies on data availability. Therefore, techniques for data enhancement play a pivotal role in preparing high-quality data inputs for GNNs. Ensuring the quality and consistency of data is paramount for accurate predictions and recommendations [73]. Consequently, this review highlights the significance of data enhancement

procedures as a critical facilitator for harnessing AI in OM applications. The aspect of system monitoring centres on the continuous collection of data from sensors, monitoring systems, and other sources, with the aim of early anomaly detection [74,75]. Performance optimisation and forecasting activities are propelled by the analysis of historical trends and current data, enabling the identification of opportunities for optimisation and the forecasting of future conditions [76,77]. These three pivotal and interconnected aspects are deemed essential for the effective management of building and civil infrastructure.

The publications were categorised based on a comprehensive analysis of their core objectives, methodologies, and outcomes, aligning with the following definitions:

1. **Data Enhancement:** Publications that focused on improving data quality, specifically through semantic enrichment by adding context to data for deeper understanding, semantic segmentation by dividing data into meaningful segments for analysis, and data quantity through imputation by filling in missing data or using a minimum number of observations to derive data to ensure completeness were considered under this category.
2. **System Monitoring:** Publications that focused on detecting anomalies that deviate from the known patterns of operation and the studies that classified the detected anomalies into known fault types were considered under this category.
3. **Performance Optimisation and Forecasting:** Publications that utilised enhanced/non-enhanced data and monitoring insights to optimise building and civil infrastructure performance and forecast future aspects were considered in this category.

3.1. Data enhancement

3.1.1. Semantic enrichment

Semantic enrichment significantly enhances the quality of data for analysis or decision-making by integrating contextual, descriptive, or categorical details, making it more informative and actionable. The objective is to augment data with additional dimensions, rather than solely addressing incomplete sections, thereby expanding its utility and enabling insights or detailed analyses that were previously inaccessible with the original dataset. When considering BIM-based data use in OM, the full potential of as-built BIM models is often underutilised due to the absence of vital FM-related information [78]. Semantic enrichment, in this context, signifies adding expert domain knowledge to BIM elements to address these deficiencies [79]. Existing approaches have evolved from using expert systems [80–82], which require a unique set of rules for each enrichment scenario, to employing advanced techniques such as Support Vector Machines (SVM) [83] and Artificial Neural Networks (ANN) [84]. However, the persistent low efficiency of existing methods, particularly those based on SVMs and ANNs, remains a significant concern [85,86]. GNNs present an alternative approach to both rule-based systems and traditional ML methods. While rule-based methods offer interpretability and domain-specific knowledge integration, they may face challenges in handling the complexity and variability of BIM data [84]. Traditional ML approaches have shown success in various classification tasks, but can struggle with the hierarchical nature of BIM data. GNNs attempt to address these issues by operating directly on the graph representation of building models, potentially enabling more efficient processing of complex element relationships and spatial arrangements. However, the effectiveness of GNNs in improving accuracy and computational efficiency for BIM semantic enrichment tasks remains an active area of research, with ongoing comparisons to established methods.

The reviewed studies that utilised GNNs for semantic enrichment primarily focused on three aspects: spatial classification, layout generation, and floor plan recommendation. The concept of “space” or “room” holds fundamental importance in FM, serving as the connecting unit between as-built BIM and OM solutions [78]. Wang et al. [32] utilised a GNN algorithm, SAGE-E to classify room types in apartment floor plans using room and connectivity features. This approach aimed to automate the labour-intensive task of labelling room types in BIM models. At present, human designers manually analyse each room’s functionality to assign the appropriate type. However, the manual graph development for the study remained a limiting factor necessitating further automation and the SAGE-E model designed for small graph datasets struggled to scale effectively for larger datasets [32]. Expanding upon this foundational research, Buruzs et al. [33] focused on automating the assignment of missing IFC entity attributes, addressing a common oversight encountered in Industry Foundation Classes (IFC) files, a widely utilised non-proprietary collaboration file format. Their efforts contributed to

the automation of spatial classification, thereby enhancing the integrity of IFC files. Given the emergence of Open-BIM concepts, solutions based on IFC files are widely embraced and regarded as a crucial preliminary requirement for the digitalisation of physical systems [87]. The SAGC-A68 dataset was further introduced to support the development of Graph Deep Learning (GDL) models for classifying space functions and elements [88]. The difficulty of explaining the prediction performance across different space function classes showcased a limitation of the utilised GAT model. The necessity for better feature integration to improve model performance further highlighted a gap in current methodologies [88].

Floor plans have traditionally served as the preferred method for stakeholder engagement, thus several studies have leveraged GNN capabilities to further automate floor plan-related aspects. Zheng and Petzold [89] introduced a subdivision GNN for room layout generation, leveraging bubble diagrams to depict the initial spatial requirements of a project. This approach simplifies stakeholder engagement during the early design stages and aids in exploring design alternatives. Their contribution extended the architecture of GCN and GAT through the incorporation of a feature fusion module. However, the challenge of generating intricate floor plans with curved walls remained unresolved. Expanding on the foundation laid by bubble diagrams, Park et al. [90] developed a floor plan recommendation system. Their SimGNN model could propose alternative floor plans based on spatial relationships. Employing a multi-layered GNN architecture, the model executed both node and graph-level embedding functions, with floor plan matches determined using a similarity score. However, with the increase in dataset sizes, the complexity of GNN models could lead to slower recommendation system speeds, negatively impacting user experience. The need for additional computational resources and further optimisation of parameters became pronounced with larger datasets. This issue was addressed by leveraging the random sampling approach during training, which highlights the ongoing challenges in handling large datasets effectively. Yang and Huang [91] demonstrated the capability to further categorise floor plans into popular, moderate, and unpopular categories by leveraging training data from end users who frequently used these buildings. The DIFFPOOL model clustered neighbouring nodes in a manner that remained largely opaque, leaving the process unexplained. The generalisability of models to other graph structures requires additional research, as existing methods prove effective only for specific types of building graphs [91]. The extended limitations of GNN applications in semantic enrichment tasks are presented in detail in Appendix G.

When implementing semantic enrichment, it is imperative to address the specific needs of diverse stakeholders. The scenarios discussed have been tailored to meet the requirements of clients, facility management personnel, and end users. Automated semantic enrichment facilitates swift adaptation to evolving organisational needs and regulatory updates, thereby reducing the need for rework, such as manual data enrichment. Semantically enriched models play a crucial role in ensuring compliance with regulations and standards [92], while also enhancing interoperability and integration with various systems, including Geographic Information Systems (GIS), asset registers, and customised operations and maintenance tools [93]. Furthermore, integrating detailed semantic information into predictive maintenance algorithms offers significant advantages by enabling precise evaluation of component conditions and performance [78], a topic explored later in this review. Despite these advancements, the creation of graph databases and their integration with GNNs often involves manual or semi-automated processes. This presents a notable research opportunity to develop more comprehensive, automated workflows tailored for facility management, thus addressing a gap in the field.

3.1.2. Semantic segmentation

Point Cloud Semantic Segmentation (PCSS) involves the automated segmentation of 3D point cloud datasets using either model-driven or

data-driven approaches. PCSS serves as a crucial component in the digitalisation of both existing and new buildings and civil infrastructure, facilitating the creation of precise 3D representations from physical structures themselves. The model-driven PCSS approach relies on pre-defined object descriptions or rulesets, which may encounter challenges when dealing with complex structures. Conversely, data-driven methods, such as GNN-based DL, leverage training data to achieve more accurate predictions [58]. Although GNN-based methods still lag behind current state-of-the-art performance metrics, they play a pivotal role in striking a balance between cost and accuracy [94]. GNN-based approaches differ from traditional methods by directly operating on the irregular structure of point clouds, potentially capturing spatial relationships more effectively. However, they may require more computational resources and complex architecture compared to established techniques such as Convolutional Neural Network (CNN) based methods. To utilise PCSS for OM, it is crucial to recognise the differences between design, construction and as-built statuses. Design data represent idealised future plans, while construction models detail construction specifics and the as-built data reflect current and past conditions with real-world imperfections. Using as-built PCSS-based models for OM facilitates the access of additional information such as precise details of wall curvatures [95], exact locations of mechanical, electrical, and plumbing (MEP) components [96], and surface textures [95]. These enhancements enable accurate maintenance planning, issue identification, and asset management.

The challenge of PCSS in diverse and complex environments that incorporate infrastructure systems has driven the development of specialised frameworks. Geng et al. [96] introduced the 3DGraphSeg framework, while Yang et al. [97] proposed the Weighted Superpoint Graph (WSPG) model. Both methods transformed extensive point clouds into a form of super-points, creating structured graphs for efficient data handling. 3DGraphSeg is tailored for high-speed railway environments, while WSPG addresses bridge geometries, outperforming traditional models such as PointNet and Dynamic Graph Convolutional Neural Network (DGCNN). The computational complexity associated with processing large-scale 3D point cloud data was a notable limitation, requiring future research to focus on optimising the efficiency of the GNN model to handle larger datasets [101]. Similarly, Le et al. [98] developed the Hierarchical DGCNN (HGCNN), and Xing et al. [99] introduced the Multi-Level and Series Pooling DGCNN (ML&SP-DGCNN) to enhance local feature extraction in their respective contexts: railway bridges and coal mining environments. Yang et al. [97] built on Superpoint Graphs while Xing et al. [99] used Dynamic Graph CNNs as the base architecture. Both studies enhanced baseline graphs to handle complexities such as class imbalance and uneven point distributions in their applications. For instance, Xing et al. [99] used multiple radius (R) values such as 0.3 m, 0.4 m, and 0.5 m to adapt to unevenness. Similarly, Geng et al. [96] proposed a two-part graph network with embedding and segmentation components to compress massive railway data. Challenges in consistently discriminating the boundaries between two adjacent components highlighted the difficulties in achieving accurate and reliable segmentation. The unavailability of Bridge Information Models (BrIMs) made it further challenging to leverage synthetic bridge point clouds [102]. When considering the use of synthetic data, using BIMs for graph-based point cloud segmentation model training is gaining traction. Chuang and Sung [100] utilised BIMs to generate synthetic point clouds, exploring the impact of noise levels on vertex classification accuracy. Shai et al. [101] extended this by using IFC model 3D geometries, observing significant performance improvement by substituting on-site point cloud data with synthetic counterparts.

The automated parsing of design data to OM platforms is fundamental in OM enhancement. The seamless transition from BIM to IFC plays a significant role in this process. Collins et al. [102] suggested that the complexity of data mapping from BIM to IFC leads to semantic information loss or alteration. Therefore, a semantic model healing methodology was proposed to address misclassified elements in IFC

models and Scan-to-BIM workflows using point cloud-based object classification. Their benchmarking study showcased the potential of GCN variants, in object classification on the IFCNetCore point cloud dataset. The performance of these GCN models, despite their lower complexity, highlights a noteworthy balance between computational efficiency and predictive accuracy. Managing deviations between as-designed and as-built models is also a widely discussed issue in the building and infrastructure OM domain. Collins et al. [103] proposed a method to bridge this gap, employing graph representations and cosine similarity metrics for similarity-based geometry retrieval. This method is established in the representation of as-designed BIM models and as-built point clouds as graphs, thereby facilitating the identification of deviations in element pairs. Consequently, this method lays the groundwork for future research aimed at automating the generation of as-built BIM from the corresponding point clouds. Collectively, these studies provide insights into the utilisation of GNNs in the semantic segmentation domain to assist in workflows such as scan-to-BIM and object misclassification, which can further strengthen the digital representation development scenarios.

GCN-based PCSS methods have also been introduced to the pavement condition monitoring domain to detect cracks from pavement point cloud scans. Through the comparison of studies conducted on similar datasets, the Saliency-based Dilated Graph Convolution Network (SD-GCN) emerged as a robust GNN architecture for crack segmentation [104]. Ma & Li [105] introduced the SD-GCN, which adopts a semi-supervised approach while incorporating additional techniques for feature enhancement. In their proposed method, Ma & Li [105] first constructed height and intensity saliency matrices to emphasise differences between crack and non-crack points. Subsequently, they employed a novel cylinder-based dilated convolution strategy to expand the receptive field of convolutions without compromising resolution. This enabled the capture of multi-scale edge features, enhancing the effectiveness of crack segmentation. However, the feature encoding performance struggled with accurately encoding features for cracks with complex structures and low connectivity, leading to misclassifications [105].

The essence of the studies within the PCSS domain highlights the significance of scenarios where precise modelling of existing buildings and infrastructure is paramount. It serves as a distinctive tool for accurately capturing and documenting geometrical features, a crucial aspect in digitalising existing projects, guiding restoration initiatives, and ensuring careful adherence to preservation standards, particularly for projects with historical significance [95]. In the aftermath of events such as earthquakes or floods, semantically segmented point clouds prove invaluable in evaluating structural damage and deformations, thereby guiding repair and reinforcement strategies [106]. Furthermore, during the construction verification process, the comparison of semantically segmented point clouds with BIM models validates adherence to design specifications, facilitating the early detection of any discrepancies [107].

3.1.3. Data imputation

Data imputation involves the replacing of missing or incomplete data within a dataset or using a minimum number of existing observations to restore its completeness. This process is crucial for ensuring that datasets are fully usable for comprehensive analysis by filling in gaps where information is absent. Traditional data imputation methods in this field have included statistical techniques such as mean/median imputation, regression-based methods, and time series approaches such as Autoregressive Integrated Moving Average (ARIMA) models [108]. At present, these methods have been developed to sophisticated DL approaches [109]. DL techniques such as LSTMs are used to capture complex temporal patterns existent in data [110]. However, GNNs differ by explicitly modelling both spatial and temporal relationships, aiming to capture the interconnected nature of networks when necessary. The ability to handle irregular data structures and varying network topologies offers an

alternative approach for data imputation when compared to other methods [111].

A recent sensor optimisation study utilised a spatial-temporal GAT-based method to reconstruct missing sensor data by leveraging reliable signals from correlated sensors installed in a cable-stayed bridge. The primary objective was to identify the most pertinent sensor data by assigning attention coefficients, thereby enhancing prediction accuracy. Attention mechanisms were employed in both spatial and temporal dimensions. The introduction of heterogeneous monitoring data into the spatial attention mechanism did not bring significant improvements as the homogeneous data already contained ample relevant information. This suggests a need for better methods to harness the unique value of heterogeneous data to enhance model performance [112]. Similarly, GNNs coupled with Graph Signal Processing (GSP) were deployed to conduct a resource-aware sensor selection process for monitoring a highway bridge. The aim was to forecast strain at sensor network nodes using the temporal-GCN algorithm. The findings consistently surpassed the benchmark across all tested conditions, highlighting the effectiveness of the GNN approach [113].

Monitoring urban water distribution systems (WDS) demands substantial resources and is often confined to specific locations. Yet, estimating flows and heads at unmonitored points is vital. Traditional physics-based methods are computationally inefficient for real-time applications, while DL-based methods, particularly GNNs, achieve efficiency through offline training. A GNN-based model demonstrated a 99 % correlation with EPANET physics-based predictions when predicting flows and heads of nodes within a 4-pipe distance from the measurement node. A key novelty was the incorporation of physical laws into the loss function during training to improve consistency. More experiments are needed to test the performance of trained models on larger networks with more complex components, such as valves and pumps, highlighting the challenge of scaling GNNs to larger and more intricate systems. The convergence issues of using least square error as the loss function and the need for exploration of other loss functions such as least absolute error and weighted least squared error points to areas for improvement in the training processes of these models. Furthermore, the study was limited to homogeneous sensors and highlights the necessity of integrating diverse sensor data [114]. Pressure estimation of WDS is also a significant task, wherein a GCN-based metamodel was employed to estimate pressure and flow rates at unmonitored stations, subsequently deriving hydraulic parameters that approximated the metamodel. Traditional hydraulic simulation models rely on equations dependent on parameters such as pipe roughness, which vary due to wear and tear. Factors such as age, material, and leakages can further impact equation-derived values; thus, GNN-based prediction methods offer valuable

insights into scenarios where physics-based methods prove impractical [34].

A similar approach was followed to enhance water demand prediction using a spatial correlation-based GCN (SCGCN) model [115], representing the sensor network in a neighbour-connected graph. Implementing a self-attention mechanism for weighting and data aggregation from neighbouring sensors, the SCGCN model outperformed ANN-based predictions. Their weighting strategy for neighbourhood aggregation is a key novelty, incorporating hydraulics-based edge weights from domain knowledge. The maintenance of Water Distribution System (WDS) databases presents a complex challenge, involving multiple parties without clear responsibility for updating the databases. A study validated a GCN-based model for predicting categorical (material) and numerical (diameter) values, surpassing traditional missing value imputation algorithms [35]. Thus, these studies underscore the potential of GNNs in various predictions related to water distribution networks. Fig. 6 depicts the trend of using GNNs for WDS-based predictions by a range of studies covering aspects such as the prediction of head, flow, pressure, demand, diameter and material. Similar to water networks, pavement monitoring also requires a substantial number of resources to carry out the monitoring throughout road networks. Traditionally missing monitoring values are either eliminated, interpolated with simple techniques, or determined by using statistical models and historical data. A collective learning-based GNN (CLGNN) [48] was proposed to impute the missing data leveraging the special structure of the road network in a GNN. However, the model had limitations in recognising nearby sections not on the same route, highlighting the need for future research to incorporate latitude/longitude information for each section. Additionally, the model did not address the interdependencies between different condition indicators, suggesting that combining all condition indicators into one multi-dimensional target could improve model performance and provide a more comprehensive understanding of pavement conditions.

3.2. System monitoring

3.2.1. Anomaly detection

The primary objective of anomaly detection is to identify data points or patterns that deviate significantly from the norm. This process often utilises unsupervised or semi-supervised learning techniques as the outliers or patterns are not previously defined. Therefore, anomaly detection algorithms are capable of flagging deviations without relying on a predefined set of fault types. The outcome is a broad identification of anomalies, which serves as an early warning system for potential issues. It is a crucial first step in system monitoring, alerting operators to

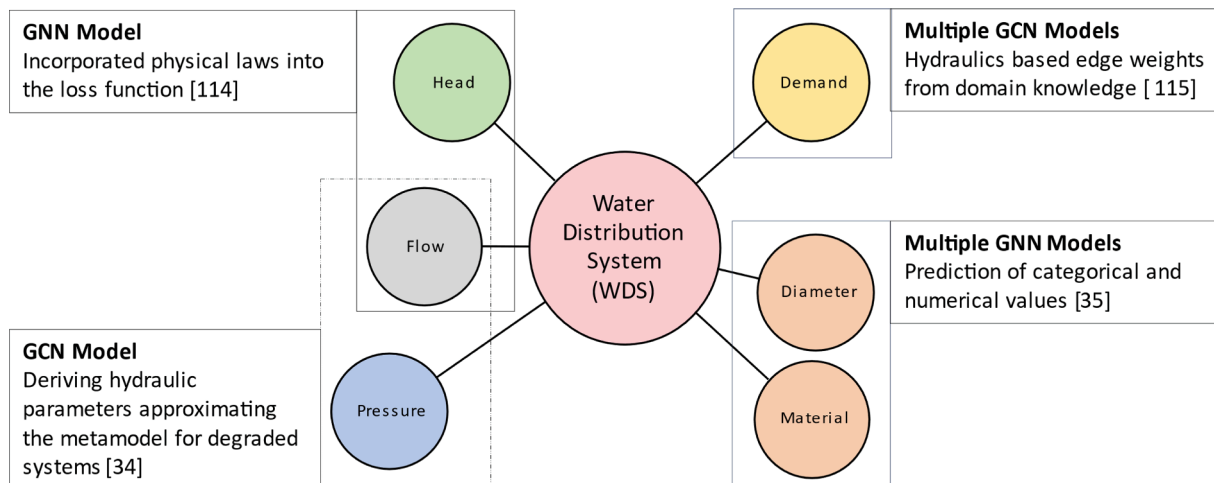


Fig. 6. GNN-based predictions for WDS.

situations that require further investigation. Existing anomaly detection approaches rely heavily on statistical methods such as control charts, moving averages, and threshold-based approaches [116]. Research studies have also been conducted in leveraging ML/DL algorithms. However, unsupervised learning techniques have gained prominence with increasing data complexity [117]. Statistical approaches typically rely on predefined thresholds, and unsupervised learning techniques may struggle with relational information. DL approaches, such as CNNs and LSTM networks have shown effectiveness in spatial and temporal anomaly detection, respectively [118]. GNNs, in contrast, attempt to integrate both spatial and temporal aspects by explicitly modelling inter-component relationships. The context-aware nature of GNNs may be particularly relevant for building and infrastructure networks, where component interactions play a significant role.

The integration of Industrial Cyber-Physical Systems (ICPSs) is essential for critical infrastructure such as water purification, oil and gas distribution, and nuclear energy plants. The identification of anomalies related to ICPS is vital for their continuous operation. An unsupervised anomaly detection system, Physics-informed Gated Recurrent Graph Attention Unit Network (PGRGA) [36], was introduced combining data-driven methods with physics-based knowledge. The model was effective in detecting anomalies in the system by modelling dependencies among variables in the ICPSs with a directed graph. The importance of prior knowledge in improving the quality of learned graph structures was also highlighted, indicating that models without such knowledge tend to underperform. Additionally, the propagation of anomalies within the system and their influence on multiple variables necessitate the use of variable causality for effective anomaly localisation. Gu et al. [119] further proposed a dynamic graph embedding approach for anomaly localisation in HVAC systems, detecting both abnormal time intervals and abnormal time series. High computational costs were a common fallback seen among these methods that model single time series or consider correlations between time series, suggesting the need for methods like LSTM to embed temporal correlations effectively. The inability to localise anomalies in advance was another critical limitation, which could be addressed by predicting sensor data to construct predictive models [119].

A similar unsupervised anomaly detection method was proposed for dam structures, using a Spatial-Temporal Variational Autoencoder (STVAE) [120]. This model captured both temporal sequences and spatial relationships within dam sensor data, integrating GNNs for spatial feature extraction and LSTM for temporal pattern recognition. The objective was to identify anomalies arising from sensor malfunctions or unexpected events, offering a comprehensive solution for detecting both temporal and spatial anomalies in dam health monitoring systems, and addressing the limitations of traditional Variational Autoencoders. However, only sensor distances were considered in the adjacency matrices, ignoring mechanical relationships. Expanding on this research direction in the civil infrastructure domain, an approach emphasised further considering spatial relations in detecting anomalies in bridge cable forces, as existing literature had primarily focused on temporal correlations. The study employed a spatial-temporal GCN, representing cable dynamometers as nodes and their spatial relationships as edges. The unsupervised model utilised data from a long-span cable-stayed bridge equipped with 168 cable dynamometers [60]. Dang et al. [121] further highlighted the concern of mislabelled training data in Structural Health Monitoring (SHM) scenarios and emphasised the approach of semi-supervised learning via the model semi-SDD. This SHM framework is categorised as a pattern recognition task, aimed to identify damage-sensitive features from collected vibration data. The study employed a deep GNN to merge spatial sensor data with temporal vibration metrics, creating potent latent representations. This approach offers a promising solution to the inherent limitations of purely supervised or unsupervised methods. The lack of mechanisms to handle the interaction between sensors impacted the overall performance. Models such as semi-SDD also face high time complexity, with significant CPU

time requirements for contrastive learning, further questioning the real-time application viability. Monitoring thermal anomalies can also be vital to understanding the behaviours of a structure. Khan and Roy [49] introduced BuiltNet, an autonomous, scalable predictive monitoring framework for analysing thermal variability in buildings. This study employed a graph-based deep clustering network to segment regions in thermal images based on temporal variations. The study identified the potential of monitoring indoor locations across seasons to pinpoint areas with significant thermal fluctuations to further take corrective actions. However, preprocessing techniques such as detrending filters and edge detection algorithms did not consistently remove noise without affecting relevant thermal data. In addition, the assignment of weight values in adjacency matrices based on temperature similarity and spatial or temporal distances did not fully capture complex thermal interactions.

3.2.2. Fault classification

Upon detecting anomalies, fault classification becomes essential to categorise these irregularities into specific types of faults based on their characteristics. The objective is to diagnose the nature of the anomaly and determine the type of fault it represents to inform the appropriate response or remediation strategy. Early approaches of fault classification relied heavily on rule-based systems and expert knowledge, where predefined thresholds and decision trees were used to categorise faults based on sensor readings and observations. The field then saw a shift towards ML techniques, with algorithms such as SVMs widely adopted for their ability to handle complex, non-linear relationships in data, while methods such as Principal Component Analysis (PCA) were used for dimensionality reduction and data preprocessing [122]. These methods improved fault classification accuracy but often required significant feature engineering. As rule-based systems rely on predefined expert knowledge, and traditional ML techniques face challenges with high-dimensional, interconnected information, advent of DL marked a major leap forward. CNNs excel at processing spatial data from images and sensors, while RNNs and LSTM networks proved effective for temporal fault patterns [123]. However, a key feature of GNNs is their ability to explicitly model the relational structure of building components, potentially enabling the capture of fault propagation patterns in interconnected systems. This context-aware approach allows GNNs to consider component interactions in fault classification, which may be beneficial in complex infrastructures. Unlike anomaly detection, fault classification typically employs supervised learning methods. It relies on a labelled dataset defining various fault types, allowing the algorithm to learn from past examples and accurately categorise new instances of faults. The outcome is a detailed diagnosis that assigns each detected anomaly to a specific fault class. This information is critical for targeted maintenance and repair activities, as it informs the operator or the system about the specific nature of the problem, enabling more precise interventions.

Building Automation Systems (BAS) enable intelligent monitoring and control of building systems, unlocking opportunities for smart, data-driven decision-making with the massive operational data that they generate [59]. Classification of this data to gain meaningful insights has gained significant attention in recent times. Thus, a method was formulated to convert tabular building operational data into association graphs, utilising GCNs to extract fault classification insights. The graph nodes symbolised entities in a chiller plant, and the edges illustrated the hierarchical and connectivity information among system components. The fault diagnosis problem was formulated as a 16-class classification problem, each featuring 3 severity levels. However, when considering model architecture improvements, increasing the number of graph convolution layers did not necessarily enhance model performance. This was due to the over-smoothing problem, where excessive graph convolutions diluted the intrinsic information of nodes [124]. Similarly, a GAT and RNN-based method [125] was proposed to diagnose faults in complex energy networks, designed to manage scenarios with missing or imperfect sensor data. This approach highlighted the GNN performance

in fault classification against traditional ML models, especially in the presence of significant data gaps. The model also enabled feature prediction alongside node state identification.

When considering fault classification tasks in civil infrastructure, Haurum et al. [126] introduced a method for sewer pipe inspection leveraging GNN-based node classification. The Cross-Task GNN (CT-GNN) decoder for multi-task classification leveraged defects, water levels, and pipe material data. The model represented defect classes as GNN nodes and modelled class co-occurrence across tasks as edges. CT-GNN surpassed existing methods for defect and water level classification on the Sewer-ML dataset which comprised of over 1.3 million images. The study highlights that imbalanced classes could lead to poor performance in rare defect classes. Adding to this research direction, Zhan et al. [127] targeted the detection of structural damages using a Deep Graph Convolutional Network (DGCN), by leveraging vibration signals. The signals were transformed into multi-sensor spatial-temporal graph-based features which were used to train the DGCN. The model leveraged GNN node classification for known damage types. The study emphasised the need for large sample sizes for accurate structural damage detection and quantification. Further, the identification of unknown damage types and locations requires advancements in algorithms. In another attempt for concrete crack detection, Residual Networks (ResNet) and GCNs were integrated [128]. Concrete surface crack features were extracted using ResNet-101 and then represented in a graph format for analysis using GCNs. This integrated approach combined ResNet's image feature extraction capabilities with graph-based techniques, resulting in enhanced image classification. The model's sensitivity to noise and missing data, necessitates further research for models that could perform reliably under various conditions. Expanding on the exploration of leveraging image-based features, a GCN prediction module was developed to identify deformation curves in structures. The objective was to integrate both local and non-local features within structures to accurately discern intricate deformation patterns. Local information pertained to adjacent feature points, uncovering localised deformations like cracks, while non-local information captured overall deformation characteristics such as settlements. This approach improved accuracy and efficiency in detecting deformations, addressing challenges associated with image quality (e.g., illumination, occlusion) and environmental factors (e.g., wind, snow, vibration) encountered by traditional methods. Additionally, the discrepancy in model performance between training and validation sets indicated the need for better generalisation ability through more comprehensive validation datasets [129].

3.3. Performance optimisation and forecasting

3.3.1. Energy-related forecasting

This category of studies aims to forecast future energy demands, optimise energy consumption, and improve the efficiency of building operations through the analysis of complex spatial-temporal relationships. This involves predictive modelling to inform design decisions and operational adjustments. Energy-related forecasting in building and civil infrastructure management has undergone significant evolution over the years. Early approaches relied heavily on simple statistical methods such as moving averages and exponential smoothing, which were used to project future energy consumption based on historical data [130]. As the field progressed, more sophisticated time series analysis techniques such as ARIMA and its variants gained popularity for their ability to capture seasonal patterns and trends in energy usage. The increasing complexity of building systems and the growing availability of data led to the adoption of ML/DL techniques [131]. DL, particularly LSTM networks, significantly improved temporal dependency modelling [132]. GNNs present a distinct approach compared to preceding methodologies, by explicitly modelling both temporal patterns and spatial relationships in energy systems. Unlike pure LSTMs, which primarily process sequential data, GNNs encode the relational structure of energy networks, potentially capturing inter-system dependencies. GNNs based methods

incorporate external influences by representing them as additional nodes in the graph structure. The application of GNNs in combination with LSTMs or GRUs further highlights a methodological distinction.

Building Energy Management Systems (BEMs) are critical elements in building management, impacting both human health and productivity by influencing thermal comfort [133]. These systems produce a vast amount of spatial and temporal data through the integration of IoT devices. However, many data-driven energy optimisation approaches that leverage this data, overlook crucial spatial interdependencies. While computational fluid dynamic models offer detailed temperature distribution insights, their complexity limits real-time application. GNNs on the other hand, can efficiently utilise both spatial and temporal data, and benefit from its offline training stage over conventional models. These applications have emerged to optimise building energy-related prediction from the early design stage to the operations stage. In the following sections, we will explore the energy-aware design approaches, cooling load prediction approaches, indoor temperature modulation approaches, and occupancy-related studies, which tend to transform traditional energy simulation practices.

As outlined in section 3.1, GNNs are used in generative floor plan design, primarily focusing on geometric attributes, and often neglect non-geometrical factors such as building energy performance. A pioneering study addressed this gap by employing GNNs to predict building energy performance from floor plan-related features represented as graphs. The study leveraged energy simulation training data generated from Radiance and Ladybug tools. Energy performance prediction was treated as a regression task, with the absence of weather data posing a limitation that could be addressed in future studies. However, manual data processing was necessary to convert design elements into a graph format, indicating the need for automated object detection algorithms [134]. Expanding from floor plan-centric to building-based predictions, the eBIM-GNN model [46] utilised prototype buildings representing typical energy usage profiles to predict city-wide energy consumption. Initially, the Point-GNN, a point cloud object detection algorithm, was employed to identify buildings from a city's point cloud, with bounding box similarity used to match the prototypes to actual buildings. This streamlined process scales energy analysis for city-wide applications without the need to remodel each building individually. This method presents a practical approach for extensive, large-scale energy analysis, potentially offering alternative energy modelling practices for smart cities or city Digital Twins (DTs). However, issues related to the granularity of class assignments and data overfitting could further complicate the effective application.

Recent studies, elaborated below, have demonstrated the potential of GNNs in enhancing cooling load predictions by considering inter-building influences, in contrast to conventional isolated building analysis. This approach is crucial as buildings in close proximity, with a flow of people in between, can influence each other's cooling requirements, creating a coupling effect. In considering above factors, a GNN model was introduced utilising spatial and temporal data to predict short-term cooling loads. The model TTGAT [50], captured spatial correlations between buildings, while its gated Temporal Convolution Network (TCN) layer accounted for dynamic cooling needs over time, outperforming traditional ML methods. Designing graph structures manually was time-consuming and labour-intensive, especially for complex and large air distribution networks. Similarly, Hu et al. [135] employed a Spatial-Temporal Graph Convolutional Network (ST-GCN) to refine urban building energy models for hourly energy use predictions. Their methodology represented individual buildings as nodes and their interdependencies, such as solar-based influences (shadows), in edges. This approach highlights the importance of capturing important inter-building relationships in energy analysis and the methods proposed are ideal for expanding BEMs to city-based DT approaches. On a further note, the impact of intra-building spatial and component relationships on cooling load prediction was also tested. Zhao et al. [51] proposed a Correlative Temporal Graph Convolutional Network (CTGCN) that

leveraged correlations among cooling system variables produced from components such as chillers, water pumps, and cooling towers over an extended period to predict cooling loads. Their method outperformed SVMs and KNeighborsRegressors. Through these advancements, GNNs have proven instrumental in recognising the spatial and temporal relationships between and within buildings and in improving the accuracy of cooling load predictions when compared to traditional ML algorithms. However, the performance of models heavily relied on the quality and quantity of training data, which was a significant limitation when high-quality data was scarce.

GNN algorithms, specifically GCNs, were employed to predict indoor temperature variations in buildings. Zones within a building were depicted as nodes, with the edges in the graph representing the heat transfer relationships between these zones. While the GCN accounted for the spatial relationships, temperature changes were captured using a LSTM network [136]. Similarly, Zhou et al. [137] proposed indoor temperature prediction by representing building zones as nodes, and walls between zones as edges in their graph representation. Contrasting from the previous study a GAT model was utilised with a GRU, which is a simplified version of the LSTM to account for temperature changes over time. However, the potential impacts of human occupancy or the operation of equipment within those zones were not considered, indicating an area for future research in building temperature prediction. Furthermore, the model's reliance on fixed graph structures limited the capturing of dynamic changes in network topology or node attributes over time [137]. Zhang et al. [138] further highlighted the limitations of traditional indoor temperature prediction methods, which often assumed uniform temperature distribution in air-conditioned spaces, disregarding spatial discrepancies. Their method of indoor temperature prediction integrated spatial-temporal data from building automation systems, IoT devices, and BIMs into a graph. The GNN, combined with an RNN (GNN-RNN), was leveraged for the prediction task. The interpretability of GNN models remains a challenge while the generalisability to different types of indoor environments is also limited as models might overfit specific characteristics of the training data.

The application of GNNs for dynamic graph structures is a prevalent trend in energy-related applications. The Active Graph Recurrent Network (Ac-GRN) and Graph Ambient Intelligence (GAIN) were two approaches that were proposed to predict district heat load, both utilising the same dataset from residential buildings in Aalborg, Denmark. The Ac-GRN model, proposed by Huang et al. [37], integrates an Active Learning Module (ALM) with a Graph Recurrent Network (GRN). The ALM actively selects the most representative heat meters, optimising the learning process by focusing on the most informative data points. The GRN captures both spatial and temporal dependencies in the data, with heat meters represented as nodes in a graph and edges determined by heat load similarity. The Ac-GRN model was evaluated over extended time horizons of 15, 30, 45, and 60 days. The substantial computational resources required to train and update the models limited their applicability and scalability in real-time and resource-constrained settings. The lack of interpretable explanations for predictions limited the applicability and trustworthiness of GNN models in real-world scenarios. [37]. Similarly, the GAIN model introduced by Wang et al. [139] employs a multi-head graph attention architecture. The Collaborative Temporal Graph Attention (CTGAT) module, a core component of GAIN, captures temporal dependencies from both heat load and meteorological observations, processed by the GAT layer, allowing nodes to dynamically focus on different neighbours. GAIN was evaluated over shorter prediction time horizons of 1, 3, 5, and 7 days. In terms of performance, the GAIN model demonstrated superior performance, especially over shorter time horizons, outperforming the Ac-GRN model when meteorological factors were considered. Scalability and computational efficiency were identified as major issues when applied to large-scale networks. Furthermore, the decision-making process of GNNs was seen as a black box, making it difficult to understand model predictions. Other limitations included the dependency on large amounts of training

data. Fig. 7 depicts the application of GNNs for building energy related prediction scenarios, from inter-room or spatial prediction to inter-building, district and city-wide predictions which scales the application in both prediction setting and impact.

3.3.2. Occupancy forecasting

The feedback collection from building occupants regarding their comfort preferences is vital for building energy system optimisation. Early approaches in understanding occupancy patterns relied on simplistic methods such as fixed schedules and basic time-of-day predictions, which were prone to inaccuracies. As sensor technology improved, infrared and CO₂ sensors became common for real-time occupancy detection. Algorithms such as SVM, Random Forests, and k-nearest neighbours (KNN) were employed to incorporate multiple variables such as weather conditions into occupancy predictions [140]. DL approaches, particularly CNNs and RNNs, introduced improvements in modelling spatial and temporal aspects of occupancy, respectively [141]. GNNs offer an approach that aims to simultaneously model the relational structure of building spaces and occupant behaviours. Unlike CNNs or RNNs, GNNs explicitly encode spatial relationships between different areas of a building, potentially enabling a range of predictions for occupant movement and space utilisation. GNNs also present a method for integrating heterogeneous data sources, such as IoT sensor inputs and computer vision data.

A study introduced the “Build2Vec” method [142], leveraging GNNs to transform a spatial graph into machine-friendly vectors, capturing the status of each building space and its interrelations. By analysing these vectors, areas with similar spatial characteristics were identified, allowing for strategic feedback collection from a variety of space types. This GNN-driven approach ensured that feedback was representative of the entire building and reduced unnecessary data collection, making the process more efficient and meaningful. When considering outdoor feedback collection, the studies on urban sidewalk comfort typically prioritise quantitative metrics, overlooking the complex relationship between individual experiences and specific environmental factors. To bridge this gap, dynamic spatial graphs were constructed using various data sources, including participant comfort scores, device measurements (e.g., heart rates, sound levels), panoramic photos, object detection results, and environmental variables, to predict outdoor comfort levels. Participants' feedback, collected through wearable sensors and smartphones, was complemented by panoramic photos captured with a GoPro camera. These data were processed, organised, and integrated into the graph structure, reflecting evolving environments and participants' experiences. Using the GraphSAGE-LSTM model, feedback was accurately predicted in the same experimental context. However, the difficulty of explaining the complex structure of GNNs was seen as a barrier to their adoption in critical applications [143]. Determination of the number of occupants in different rooms at varying times is another crucial factor for energy simulations. Traditional methodologies often rely on static schedules, which do not mirror real-world scenarios. The capability of Graph-LSTM was tested to predict the simulations that were generated by the agent-based and discrete event modelling software AnyLogic. This involved simulating 5 distinct office layouts over 200 workdays. RNN, GRU and LSTM baseline models predicted the regular occupancy patterns accurately, while GRAPH-LSTM was capable of predicting random events as well. The complexity of GNN models introduced unwanted noise, making it challenging to manage and interpret the results accurately [144].

4. Implementation, limitations, and future directions

Following the review of the applications in section 3, an implementation workflow for GNN-based analysis was established (Fig. 8) referring to the best practices observed. The overall process is divided into subprocesses of graph databases, representation, optimisation, application, supporting technologies, and implementation and

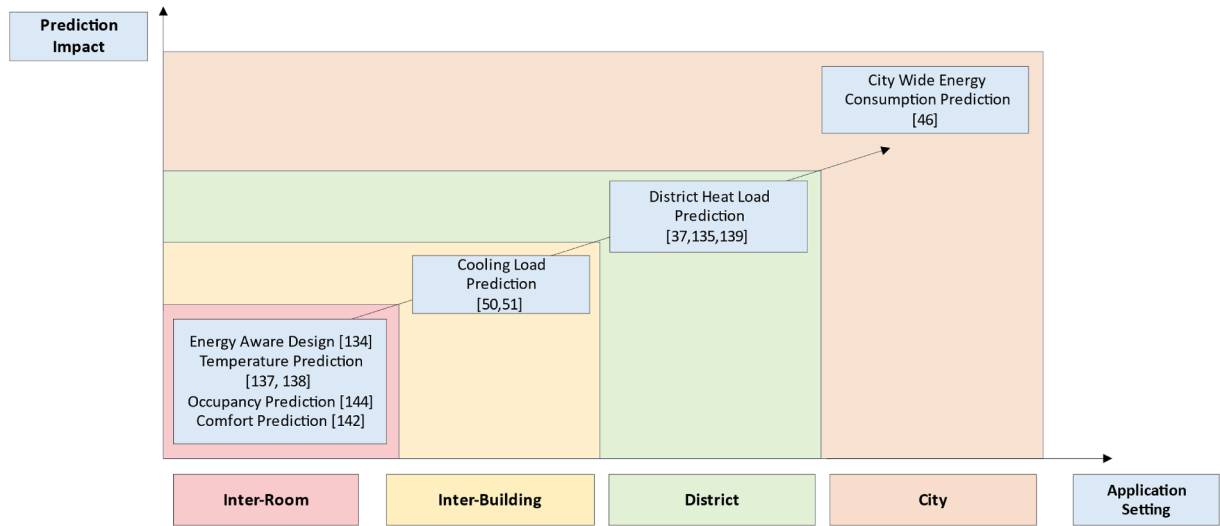


Fig. 7. Scaling up energy prediction scenarios with GNNs.

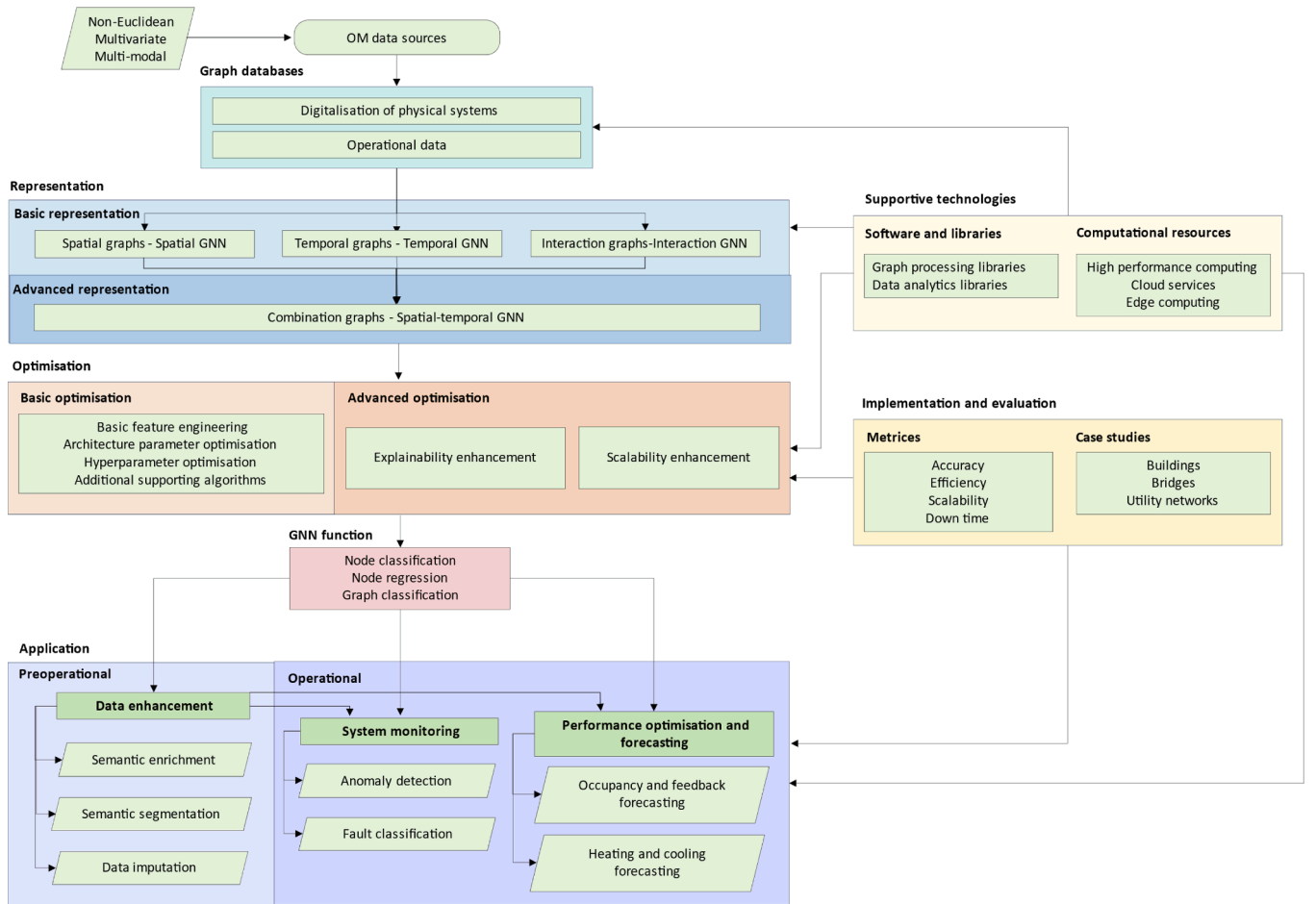


Fig. 8. Implementation workflow for GNN-based analysis.

evaluation. The summary of the process is presented here, while specific aspects will be discussed in detail in the subsequent sections. The process is initiated by utilising OM data sources supplemented by their physical representations, which include BIM data, IoT sensor data, maintenance records, operational data, etc. Initiating the representation aspect, raw OM data is subjected to preprocessing to transform into suitable graph formats for analysis. Various types of GNNs such as GCN,

GATs, and GraphSAGE are utilised for different aspects of the graph data, categorised by spatial, temporal and interaction characteristics. Combination graphs are processed using advanced spatial-temporal GNNs.

In the optimisation stage, the GNN architecture is enhanced through basic optimisation techniques, which include the modification of architecture and hyperparameters, as well as feature engineering aspects.

Advanced optimisations such as explainability enhancement and scalability enhancement are also performed which will be further discussed in sections 4.4 and 4.5 respectively. The prominent GNN functions utilised are node classification, node regression, and graph classification. The application stage includes GNN-based data enhancement scenarios such as semantic enrichment, semantic segmentation, and data imputation to improve the quality and quantity of the data. Either enhanced or non-enhanced data is then used in applications categorised into system monitoring, performance optimisation and forecasting. Supportive technologies are also used to aid in the implementation and evaluation. The performance of these applications is measured from the perspective of OM enhancement through various metrics such as accuracy, efficiency, scalability, and downtime apart from other DL-based accuracy determination matrices. Case studies on buildings and civil infrastructure systems are further conducted to evaluate the practical implications and to understand their impact on real-world scenarios.

4.1. Limitations and future directions

As highlighted in Table 1, current semantic enrichment applications are largely focused on interpreting and optimising floor plans. However, the potential shown by GNNs in recognising extended spatial features of these floor plan representations is noteworthy. In semantic segmentation, the improvement of the accuracy of BIM models and the facilitation of the transition from physical scans to digital representations is achieved through the utilisation of GNNs. The domain of missing data imputation is currently limited to water networks, structural health monitoring systems, and road networks. Further limited are the applications in anomaly detection, which are confined to cyber-physical systems, structural health monitoring, and building thermal monitoring. The utilisation of GNNs in system monitoring is illustrated through the classification of faults in energy systems, image-based diagnostics, and sensor-driven fault detection. Additionally, the scope of

Table 1

GNN utilisation in the building and civil infrastructure OM domain.

Category	Objective and references	Models/Modules	Key nodes	Key hyperparameter range
Data Enhancement				
Semantic Enrichment	Enrich space function information [32,33,88]	GraphSAGE, GCN, GAT, MLP	Spaces/Rooms labels	Layers: 1–8 Batch Size: 1–4 Learning Rate: 0.0001–0.01 Epochs: up to 10,000
	Floor plan generation, classification, and recommendation [89–91]	GraphSAGE, DIFFPOOL GNN, Subdivision GNN, GCN, set2set, hybrid geometry optimisation	Spaces/Rooms	GRU/GCN Layers: 2–3 Batch Size: 4–8 Learning Rate: 0.001–0.005 Epochs: up to 10,000 Embedding Size: 10–128
Semantic Segmentation	As-built model improvement using point clouds [102,103,145]	GCN, BIM2Graph module, ifc2vec, cosine similarity	Building elements, Points	
	Synthetic point clouds for Scan-to-BIM enhancements [100,101]	GCN, GRU, GRAN	Points, Superpoints	
	Point cloud segmentation improvement for Scan to BIM [96–99]	GCN, DGCNN, GRU, PointNet	Points, Superpoints	
	Pavement crack segmentation [104,105,146]	GCN, GraphSAGE, MLP	Point, Pavement sections	
Missing Value Imputation	Water distribution systems [34,35,47,114,115]	GCN, GAT, ChebNet, GraphSAGE, and TAGCN	Junctions, Manholes, Sensors	GCN/GNN Layers: 2–4 Learning Rate: 0.001–0.01 Batch Size: 16–3000 LSTM Layers: 1–2 Hidden Units: 20–256
	Structural health monitoring systems [112,113]	GAT, GCN, Spatio-Temporal Graph Convolutional Network (T-GCN), Graph Signal Processing (GSP), and LSTM	Sensors	
	Road networks [48]	GraphSAGE	Pavement Sections	
System Monitoring				
Anomaly Detection	Industrial cyber physical systems [36,119]	GRU, GCN, GAT	Sensors, Actuators	Batch Size: 30–256 Learning Rate: 0.001–0.01 Epochs: 100–300 Dropout Rate: ~0.2
	Structural health monitoring systems [60,120,121]	1DCNN, Graph Transformer, GCN, RNN, Variational Autoencoder	Sensors, Raw time-series vibration data, Time series of cable force measurements	
	Thermal anomaly detection [49]	GCN, LSTM	Regions in thermal images	
Fault Classification	Energy system fault detection [124,125]	GCN, FCNN, Cheb, SAGE, GAT, GCN2, GAT, GRU, and GRNN	Data samples, Sensors	GCN Layers: 1–4 Learning Rate: 0.0001–0.1 Epochs: up to 500 Dropout Rate: ~0.2 Hidden Units: varied Kernel Size: varied
	Image-based fault detection optimisation [126,128,129]	GCN, GAT, ResNet-101, and hybrid residual networks	Class embeddings across tasks, crack feature maps from images	
	Sensor based fault classification [127]	Continuous Wavelet Transform (CWT), DGCN, and spectral GCN	Multisensor spatial-temporal graph-based feature vectors derived from CWT coefficients	
Performance Optimization and Forecasting				
Occupancy and Feedback Forecasting	Building occupancy and sensory feedback prediction [142–144]	Hybrid GNN, LSTM, GraphSAGE, Build2Vec	Spaces/Rooms, Environmental Variables, Virtual Nodes	Layers: 5, Epochs: 100–200, Learning Rate: 0.0001–0.001 Loss Function: Cross-entropy
Heating and Cooling Forecasting	Predicting floor plan/BIM model-related energy performance [134,147]	GraphSAGE, GCN, GatedGCN, GAT, MoNet, Point-GNN, MLP	Spaces /Rooms, Buildings	Layers: 3–16 Learning Rate: 0.001 Epochs: ≤1000 Hidden Size/Units: 24–128
	Indoor temperature prediction [137,138]	GNN, GAT, DL-GCN, LSTM, GRU, RNN	Zones, Physical entities (VAV boxes and sensors)	
	Building cooling load prediction [50,51]	GCN, TTGAT, GTC	Variables influencing cooling load	
	District heat prediction [37,135,139]	GCN, ST-GCN, GAT, GRU	Clusters of heat load observations, Heat meters, Individual buildings	

GNN applications within the optimisation and forecasting domain is primarily focused on energy-related studies, demonstrated by their specific use in analysing occupancy patterns, enhancing energy efficiency through performance forecasts, and predicting temperature fluctuations. In addition to the application scope limitations, further technical limitations were also discovered. These limitations are summarised below in Table 2.

In summary, the potential of GNNs in enhancing the operation and maintenance of building and civil infrastructure is showcased by the addressed scenarios, yet the emphasis on these specific applications reveals a gap in addressing a broader spectrum of challenges. The implementation, feature engineering and basic optimisation driven by the other limitations as per Table 2 will be discussed in section 4.2. However, the expansion of the application of GNNs to include further data interoperability, explainability, and scalability could offer comprehensive cross-system applications of the above standalone tasks. Fig. 9 summarises highlights from section 4, in which the challenges and future directions are represented as an extension to the previous workflow diagram represented in Fig. 8. The diagram highlights the stages of graph data generation, advanced optimisation techniques, and future application scenarios that will be further discussed in subsequent sections, exploring innovative research directions undertaken to address them based on insights from the reviewed studies.

4.2. Implementation, feature engineering and basic optimisation

The surveyed studies span across a diverse array of GNN applications in the building and civil infrastructure OM sector, as comprehensively detailed in Table 1. This table highlights the extensive predictive capabilities of GNNs, with a notable variation in the feature utilisation and complexity of the algorithms employed across different use cases. Additionally, for an in-depth understanding, extended tables encompassing, node and edge features, GNN architecture parameters, and optimum hyperparameters per study are included in the Appendix B, Appendix C, Appendix D, Appendix E, Appendix F, Appendix G arranged by sub-categories. Fundamental GNN architecture, such as GCN, GAT, and GraphSAGE were widely used with case-specific modifications which are documented in detail in these appendices, thus future studies can leverage the GNN architecture parameters and hyperparameters as an initiation point for similar analysis development. Although similar algorithms are widely utilised, unique functionalities of GNNs and feature engineering improvements have provided the means to produce distinctive results. Thus, the feature engineering process holds an important role in GNN-based analysis.

Given that the features used in each study are detailed in Appendix B, Appendix C, Appendix D, Appendix E, Appendix F, Appendix G for further reference, it is worth exploring the trends in feature engineering, as demonstrated by a building energy management system-based energy

consumption forecasting scenario outlined in Table 3. Here, the nodes serve as data sources representing distinct components within a campus setting, including HVAC units, rooms, and entire buildings [50,139,148]. Subsequent steps in the process involve the determination of additional attributes corresponding to these nodes and their connecting edges. Notably, the nodes encapsulate key features such as HVAC unit efficiency, energy consumption or generation, room occupancy levels, and building material thermal properties. In contrast, edges play a crucial role in integrating connections or interactions. Although basic GNN architecture lacks the inherent capability to utilise edge features, minor adjustments have facilitated such integration [32]. Edges are instrumental in encoding various types of physical connections or interactions, as evidenced by ductwork connections between rooms and HVAC units. Moreover, edges can convey additional information such as connection strength, capacity, or efficiency. These features are incorporated into GNN training through matrices, primarily represented by numerical values.

Ensuring the proper encoding of features is paramount for facilitating the model's comprehension and processing. For categorical data such as HVAC system types, employing one-hot encoding is imperative, as it delineates these categories as distinct entities without presuming any ordinal relationship among them. Meanwhile, utilising denser embeddings for intricate data sets, such as occupancy schedules, enables the model to capture complex patterns effectively. In addition to direct data representation, derived features play a pivotal role, manifesting as new attributes or properties generated based on the original features of the nodes or edges. Noteworthy examples include the temporal and spatial coupling of cooling loads across different buildings, correlations among weather factors, and correlations among buildings [50]. Moreover, global graph features encapsulate the principal characteristics of the system under consideration. Exploring further, analyses pointed in on subsystem properties, such as node centrality, are encoded into GNNs using features. An illustrative application is the identification of critical energy consumption areas through subgraph isolation based on specific features [138]. The identification of sub-graphs is a unique scenario specific to GNNs that eases the incorporation of subsystem features into an analysis.

While an abundance of features may be available, the precise selection of relevant ones is imperative for attaining optimal outcomes. The significance of feature engineering cannot be overstated, as it plays a crucial role in mitigating computational burdens and enhancing efficiency. To identify the most pertinent features, researchers have turned to gradient-based methodologies, such as saliency maps. These techniques involve computing the gradient of the output concerning each input feature, thereby enabling the identification of key features characterised by higher gradient values [105]. GNNExplainer [149] shows potential as a specialised method to further understand GNN-based analysis, exhibiting the potential to accurately discern specific

Table 2
Summary of GNN-based application limitations.

Limitation Type	Description	Study References
Data Interoperability	Inconsistent data formats across different datasets require extensive manual annotation and preprocessing efforts, complicating data integration and analysis. The lack of standardisation hinders the creation of comprehensive datasets, leading to challenges in generating synthetic data and extracting meaningful features from complex data structures.	[89,32,99,97,112,120,49,125,127,50,51]
Scalability	Models often face difficulties in handling large datasets efficiently, resulting in high computational complexity and significant resource requirements. This includes issues such as increased memory usage, slower processing times, and the need for specialised hardware or optimisation techniques. Fixed-size neighbourhoods in some models limit their adaptability to varying data scales, posing challenges for scalable deployment.	[90,32,101,96,35,114,119,121,139,37,143]
Explainability	The internal workings of many models remain opaque, making it difficult to interpret and trust their outputs. This lack of transparency can lead to variability in prediction performance and challenges in understanding decision-making processes.	[91,88,34,48,126,139,37,143,138]
Other Limitations	Issues such as class imbalance, insufficient feature integration, and limited generalisability to different data structures affect model performance and applicability. Specific theoretical and practical limitations, including difficulties in handling complex boundaries and accurately capturing spatial design features, further restrict the utility of current methodologies.	[88,89,91,146,105,48,114,124,129,142]

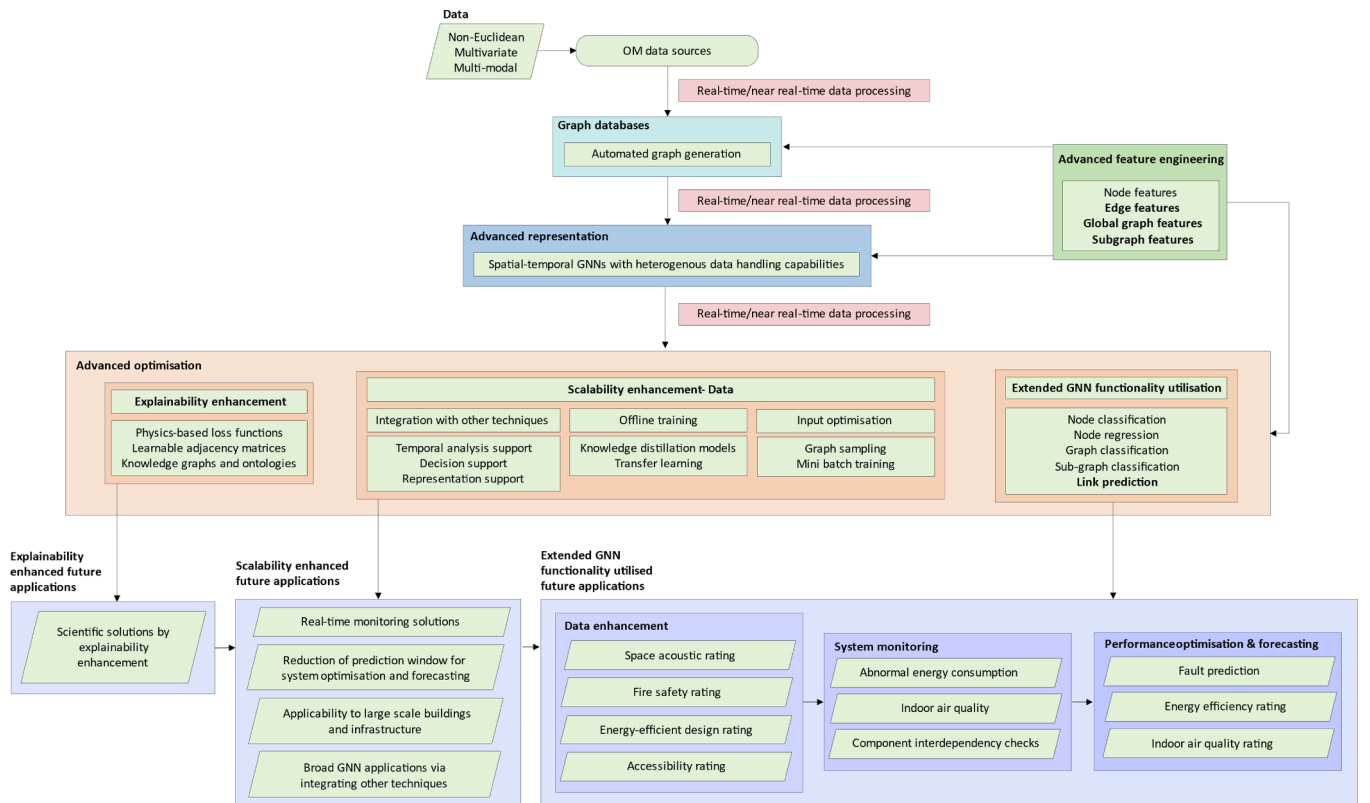


Fig. 9. Concept map of the discussion, challenges and future directions.

Table 3
Feature engineering aspects relevant to a building energy management system.

Aspect	Description
Component Nodes	Identifying the primary elements of the system to be modelled as nodes in the graph, which are central to the analysis – HVAC units, rooms, or buildings
Edge Relationships	Establish connections between nodes based on physical/logical interactions/correlations – Ductwork connections between rooms and HVAC units, ductwork efficiency
Feature Selection	Choosing attributes of the nodes/edges that significantly influence the system's behaviour or performance – thermal properties, HVAC efficiency, room occupancy
Feature Encoding	Transforming the selected features into a format that the model can effectively process – one hot encoding for HVAC system types and denser embeddings for occupancy schedules
Global graph features	Overall system characteristics – total energy use, component efficiency, energy source diversity
Subgraph features	Subsystem properties – key energy-consuming areas

subgraphs and node features that apply significant influence over the model's predictions. By maximising mutual information, a measure indicative of the extent of knowledge gained on one variable through another, GNNExplainer facilitates the pinpointing of exact graph elements and attributes pivotal to the model's outcomes. Furthermore, Layer-wise Relevance Propagation (LRP) [61,150] also offers another viable solution to identify important features by backpropagating the output through the network to assign relevance scores to the input, thereby informing the significance of individual features in driving model performance.

Further advancement in harnessing GNN capabilities can be achieved through the optimisation of existing GNN architecture. Many publicly available models offer allowances for modification, as evidenced by the development of the SAGE-E algorithm, which represents a well-documented adaptation and refinement of the original GraphSAGE framework [32]. Expanding the number of layers within a GNN

algorithm stands out as a fundamental optimisation strategy aimed at capturing intricate interactions. However, it is crucial to acknowledge the inherent risk of overfitting associated with such enhancements. Overfitting in GNNs occurs when the model excessively tailors itself to the training data, capturing noise and outliers rather than discerning underlying patterns. Techniques such as graph dropout [90], and residual [134] and skip connections [50] have been leveraged in several studies to prevent such overfitting. Graph dropout randomly removes edges or nodes to promote generalisation; residual connections facilitate deeper model training by allowing gradients to flow through layers more effectively, directly connecting input from a previous layer to a subsequent layer's output. Skip connections, similar to residual connections, enable the model to learn an identity function, ensuring that deeper layers can perform at least as well as shallower ones, thereby improving the model's ability to generalise to unseen data while mitigating the vanishing gradient problem. These optimisation techniques can be considered as some of the most common modifications seen among the studies in this domain.

4.3. Data interoperability enhancement future applications

The stage of data preprocessing for GNN-based analysis involves the collection, normalisation, and transformation of data from diverse sources such as building plans, 3D models, IoT sensors, maintenance logs, etc. This stage is crucial for standardising the data format, ensuring consistency across different data sets, and identifying the entities (spatial, temporal, interaction, combination) and their relationships that will form the nodes and edges of the graph. If the graph representation fails to capture the essential relationships within the data, GNNs' performance can significantly drop. When considering these vivid data sources, the data interoperability for the current context is defined as the ease of using data from raw sources in a GNN algorithm. As GNNs predominantly focuses on graph-based data, every data source needs to be converted to a graph representation followed by a matrix

representation. A data enhancement stage can also be incorporated, where a GNN algorithm itself can be leveraged to enhance the obtained data using semantic enrichment, semantic segmentation or data imputation as discussed in [section 3](#).

The majority of the studies that were considered in this review utilised manual approaches for graph generation. However, applications relevant to the semantic enrichment category have shown a trend in progressing towards automated graph database generation. Ziaee and Suter [88] made a significant contribution with the SAGC-A68 space access graph dataset, leveraging methods by Suter [151] for space data extraction from BIM. In addition, although accounts for a different phase of a project, the studies from the construction domain have shown progress in generating graph databases, which can be further utilised for OM workflows. This is exemplified by the studies from Wang et al. [152] who developed multi-disciplinary BIM model-based graphs to address design inconsistencies; Hu et al. [153] and Hu et al. [154] who introduced clash representation graphs; Gan [155], who explored modular building graphs; Zhu et al. [139], who explored IFC-based graphs to enhance information access. A further research direction exists in identifying methods to completely automate this process of graph generation and further direct integration to GNN analysis.

When considering graph data types, homogeneous graphs are characterised by nodes of uniform type and edges denoting consistent types of relationships. This simplicity allows for the straightforward application of GNNs, as the model can learn to propagate information through the graph's structure without needing to differentiate between different types of nodes or edges. Heterogeneous data, on the other hand, involves graphs containing multiple types of nodes and/or edges. These graphs are more complex, reflecting a richer variety of entities and interactions. Heterogeneous graphs require more sophisticated GNN models that can handle this diversity, such as models that incorporate different relation types explicitly or models that learn to aggregate information differently depending on the node and edge types, such as Heterogeneous GNNs (HeteroGNNs) [65]. HeteroGNNs employ diverse aggregation schemes tailored to varying relationship types, or they may integrate *meta*-path-based strategies to gain insights from intricate patterns of interactions spanning different node and edge types. Studies such as Bloemheuveld et al. [113] and Xiong et al. [156] showcase the effectiveness of integrating such heterogeneous data. Bloemheuveld et al. [113]'s work in SHM demonstrates how temperature data can enhance the accuracy of missing cable force data restoration. Thus, further research on the integration of heterogeneous data types is encouraged.

Real-time or near real-time data processing and analysis is crucial for some of the considered applications in this review [36,50,137]. This capability is particularly important in creating advanced systems derived from the aspects discussed in [section 3](#). Research into early warning systems for infrastructure failure is a promising direction that can leverage real-time sensor data. This approach builds on existing studies in dam anomaly detection [120] and bridge anomaly detection [60]. In other research directions, the energy demand forecasting prediction window can be reduced by utilising real-time data [51]. Real-time integration ensures that the models are working with the most current data, enabling timely decision-making and interventions that can prevent failures and optimise resource use. Future research in the domain also could focus on creating robust mechanisms for dynamic update of GNNs. One promising approach is to employ event-driven update mechanisms where a GNN model is updated in response to specific events [157]. This approach is more efficient than constant updates, as it only activates changes in the model when significant alterations in the graph structure are detected. GNNs further possess the unique capability of augmenting existing graph structures by introducing additional nodes and edges, without necessitating complete retraining from scratch. This versatility stems from their node and edge-based architecture [158]. Utilisation of such approaches ensures that the models remain up-to-date and reflective of the latest data, enhancing their accuracy and efficiency in dynamic environments [159].

4.4. Explainability enhancement future applications

The advancing research trend of Explainable AI (XAI) refers to the ability to understand and interpret the decisions or predictions made by AI models. When considering applications relevant to building and civil infrastructure, black box models are discouraged as the domain deals with aspects relevant to human lives [160]. Enhancing explainability improves stakeholder trust and facilitates the implementation of GNN recommendations for real-world applications. Explainability also aids in diagnosing and correcting model biases or errors, further refining the system's accuracy and reliability. A trend in focusing on explainability enhancement has been identified from studies such as district heat forecasting [37].

In enhancing the explainability, integrating domain knowledge at different stages of implementation has been identified as a viable solution. The initial representation of the data structure in a domain-appropriate graph as seen in various studies relevant to WDS and road networks significantly impacts model performance and interpretability [34,48,114,115]. A case study by Belghaddar et al. [35] focusing on wastewater network data completion illustrates the efficacy of incorporating domain expertise throughout the modelling process. The inclusion of Strahler's number as a feature to represent downstream pipe diameter increments reflects an understanding of network topology that may not be apparent from raw data alone. Domain expertise also plays a crucial role in variable selection and addressing data imbalances. The study further highlights how knowledge of pipe diameter-material correlations informed feature selection, while an understanding of the relative importance of different pipe classes guided the treatment of class imbalances in the dataset. This approach ensures that the model focuses on adequately represented and practically significant classes within the domain. The selection of evaluation metrics can be further tailored to align with field-specific priorities when guided by domain knowledge. In the same study, emphasis was placed on recall for minority pipe classes, reflecting the practical importance of accurately identifying these less common but potentially critical components of the system. Moreover, domain knowledge facilitates the interpretation of results within the context of real-world systems. The variation in Strahler's numbers across different network sizes can be explained through an understanding of urban wastewater system structures, providing insights that extend beyond mere statistical correlations.

When further considering the application of domain knowledge to DL algorithms in engineering, Physics-Informed GNNs (PIGNNs) are among the emerging but less utilised methods. As demonstrated by Wu et al. [36] and Zhou et al. [137] incorporating physical principles into the GNN structure and loss function enables accurate modelling of complex systems ([Fig. 10](#)). To integrate physics-based loss into a GNN, a loss function is defined such that it quantifies the discrepancy between the predictions made by the GNN and the known physical laws or principles governing the system. The network is trained on this combined physics-informed and data-driven loss. PIGNNs are particularly effective in handling uncertainties in large or dynamic systems with sparse or noisy data. Shu et al. [120] further acknowledge this methodology in their Sequential Variational Autoencoder (SVAE) designed for anomaly detection. They incorporate additional loss terms to enforce temporal smoothness and induce similarity, ensuring spatial consistency among sensors. The utilisation of PIGNNs shows promise in SHM applications, facilitating the seamless integration of stress and strain relationships within structural components using elasticity theory as the basis for the loss function. The building energy efficiency optimisation can be further modelled using heat equation-based loss functions, and the pipe flow values in utility networks can be described by concepts of head loss and mass conservation. Thus, PIGNNs have vast potential in the building and civil infrastructure domain, a research direction only addressed by a few studies.

Dynamic adjacency and learnable adjacency matrices are also concepts identified to improve the explainability of GNNs. According to Li,

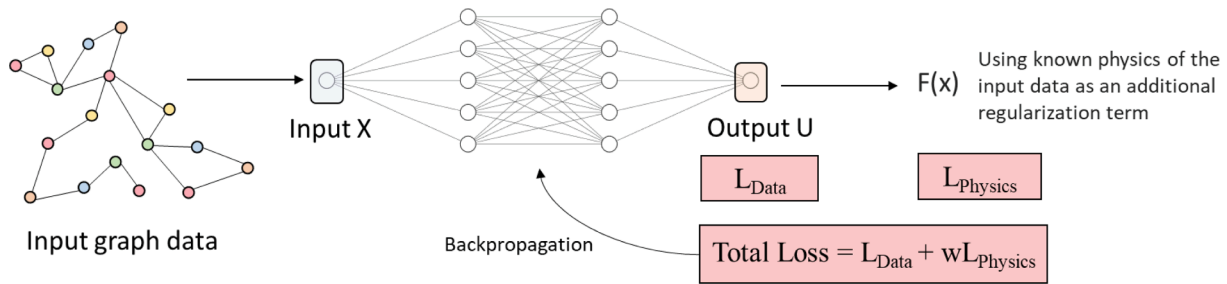


Fig. 10. Physics informed GNN architecture.

Niu and Li. [112], a learnable adjacency matrix in a GCN is one that is automatically learned during the model's training. This matrix was employed in spatiotemporal GCN models to capture spatial dependencies between vertices (sensors), facilitating the identification of latent correlations and spatial dependencies through training. This approach offers increased flexibility and accuracy for spatiotemporal modelling. Zhao et al. [51] further introduced a data-driven method to learn graph structures by analysing correlations between key variables in cooling load prediction. These correlations were used to construct an adjacency matrix that represented the weighted connections between nodes. The importance of these dynamic matrices lies in the opportunity presented to change the matrix structure, mid-problem, rather than depending on predefined static matrices.

Knowledge Graphs (KGs) and ontologies are powerful tools for enhancing the explainability of GNN-based analysis, especially in complex domains. The explicit representation of entities and relationships relevant to an engineering problem by KGs allows stakeholders to understand the basis of the GNN predictions [161,162]. Specific ontologies have been identified for domains such as semantic enrichment, where ifcOWL plays an important role in defining the BIM graph structures to be utilised with GNNs. Further integration of ontologies such as BRICK, Sensor, Observation, Sample, and Actuator (SOSA), Semantic Sensor Network (SSN) and Building Topology Ontology (BOT) [163] can enhance applications such as fault prognosis, which is yet to be seen as a prominent research direction in this domain. These facilitate the integration of diverse data sources, such as sensor data, maintenance logs, and weather information, into a unified framework. KGs can further assist in tracing the predictions back to root causes, in potential failure detection scenarios such as HVAC system failures. In addition, the link prediction capabilities of GNNs can be applied to KGs in predicting unforeseen knowledge links which can further assist in prognostics [164]. Therefore, KGs provide a second layer of validation that adds to the model's credibility in validating the GNN predictions against known constraints and rules in the building and civil infrastructure domain [165]. This integrated approach enables GNNs to leverage a richer context for analysis, improving the accuracy and reliability of predictions [166].

4.5. Scalability enhancement future applications

Scalability in data handling is crucial for GNN models to manage the growing complexity and size of building and infrastructure networks. This capability supports tasks such as comprehensive risk management, life cycle analysis, and resource optimisation across large-scale networks, including those used in utility management and city-wide infrastructure systems. The scaling up encompasses various techniques that are focussed on predominantly optimising the input and also the algorithms. One such input optimisation approach identified from the review is graph sampling, which employs techniques such as node sampling, layer sampling, or subgraph sampling to reduce the computational load by training the GNN on a strategically chosen subset of the graph [32]. Another method identified is mini-batch training, where GNNs are trained using mini-batches of nodes or subgraphs, thereby

significantly reducing memory requirements [32]. The mini-batch training approach by Wang et al. [32] demonstrates a specific use case where it enhances the efficiency of the GraphSAGE algorithm for BIM data, by combining numerous small graphs into a large disjoint meta-graph. This allows for parallel processing and efficient convolutions in GNN layers, greatly improving training and inference efficiency. This also involves parallelising the computation across multiple GPUs or distributing it across clusters, greatly speeding up the training process for large-scale graphs. Tools such as Deep Graph Library (DGL) or PyTorch Geometric were vastly utilised in facilitating efficient parallelisation and distributed training. These methods strike a balance between accuracy and efficiency, proving advantageous in scenarios demanding real-time processing. However, it is essential to utilise further advanced computing technologies such as edge computing, and cloud computing as supporting technologies to scale up GNN applications for real-time analysis capabilities.

In considering further scalability scenarios, GNNs can be deployed on a wide range of OM data analytic systems by training them on historical offline data. This is particularly beneficial for tasks such as anomaly detection [36]. The proposed methods for offline training include knowledge distillation models, also known as student-teacher models. In this approach, a simpler 'student' model learns to mimic a more complex 'teacher' model, allowing for deployment in resource-constrained environments while maintaining high performance as represented in Fig. 11. This method is instrumental in training GNNs to handle raw graph data and generalise better across different graphs or subgraphs, making them scalable for large and complex infrastructure systems [167]. Transfer learning [168] can be proposed for scenarios where a model developed for a specific task serves as the foundation for a model on a second task. The availability of pretrained models on similar tasks or datasets could be used to transform raw data into a feature space that the model has learned to be effective in. These techniques are particularly valid for complex data types such as time series or textual information (maintenance logs). The significance of this approach is further highlighted in GNN-based DL, as it enables training with relatively less data, proving advantageous in domains where abundant training data is scarce.

It is crucial to acknowledge the limitations of conventional GNNs in scenarios such as time series data analysis, decision-based analysis, and multi-hop neighbourhood analysis. Complementary algorithms could be proposed to be integrated with GNNs to fulfill supporting roles in these contexts, often added as additional layers as represented in Table 4.

As detailed in Table 4, and thoroughly discussed in Section 3, LSTMs or GRUs are seen as prominent algorithms to process the temporal aspect, while the spatial relationships have been captured by GNNs [136,138,144]. In addition, integration of decision support algorithms can be identified as another possible research direction. Reinforcement Learning (RL) algorithms harness the relational structure captured by GNNs to acquire optimal decision-making policies. They achieve this by engaging with the entire environment relevant to the problem at hand and obtaining feedback in the form of rewards [169]. Optimisation algorithms on the other hand show promise in scenarios of seeking the best outcomes under given constraints, exemplified by devising cost-

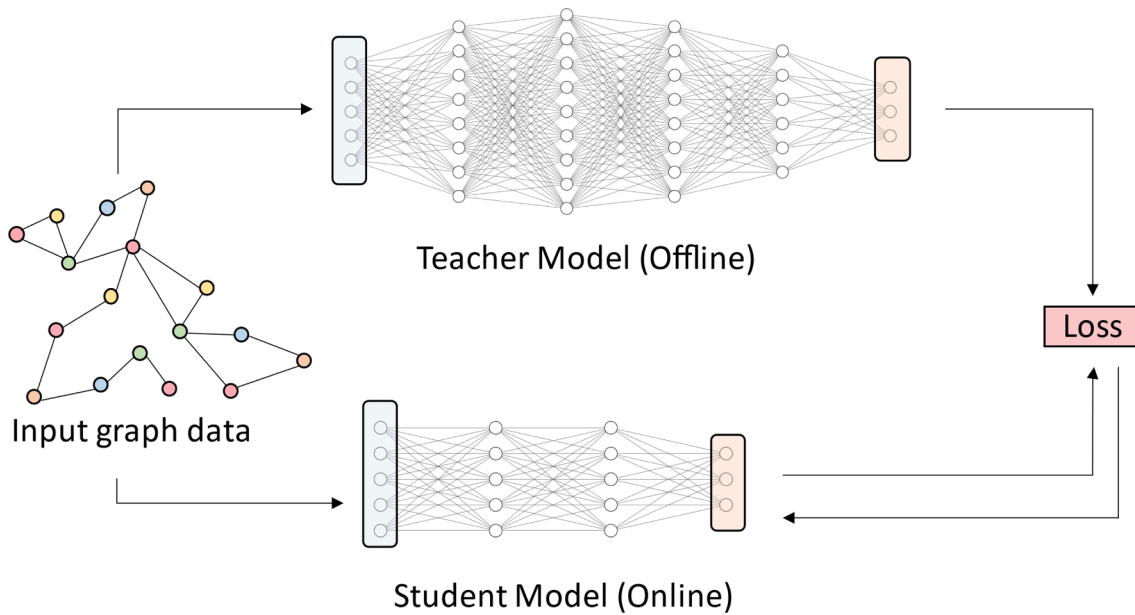


Fig. 11. Overview of a knowledge distillation model.

Table 4

Incorporation of supporting algorithms to scale up GNN functionalities in the building and civil infrastructure OM domain.

Technique	Function delivered by the additional technique
Proposed temporal support algorithms	
LSTMs/GRUs [136,138,144]	Capturing and leveraging temporal patterns in data sequences
Proposed decision support algorithms	
Reinforcement Learning [169]	Optimised decisions through trial and error in dynamic environments
Optimisation Algorithms [170]	Identifying the best solution under given constraints
Bayesian Networks [171–173]	Inferring probabilities and making decisions under uncertainty
Proposed representation support algorithms	
Graph Autoencoders [120]	Encoding data into compact representations for reconstruction or feature extraction
Graph Transformers [174]	Capturing long range dependencies

effective schedules to enhance maintenance planning [170]. The combination of GNNs and Bayesian networks can be further considered for risk assessment and decision-making under uncertainty where probabilistic relationships exist [171–173].

Furthermore, several other algorithms have also shown promise in providing a computational assistance role. The compression of high-dimensional input features, such as occupancy schedules or thermal properties into a lower-dimensional, dense representation has been facilitated by the use of autoencoders, which was exemplified by the study on dam anomaly detection [120]. This process involves the separate training of the autoencoder on the dataset to learn an efficient encoding of the input data, which can later be utilised as input for the GNN. Alternatively, an autoencoder can be directly embedded into the GNN architecture, where component attributes are first passed through an encoder to reduce dimensionality, and then these compressed features are utilised within the GNN layers for node representation learning. As understood throughout the review, GNNs excel in leveraging the localised structural relationships between components, efficiently processing information from immediate neighbours to which systems are directly connected. Graph Transformers, a novel development that is underutilised in this domain, would instead focus on capturing the global interactions across the entire building operation

system, irrespective of the components' physical proximity, by applying self-attention mechanisms, which can identify complex, long-range dependencies and interactions between systems [174]. This capability could be particularly useful in optimising energy usage and operational efficiency on a larger scale or in complex scenarios where indirect interactions significantly impact performance and maintenance schedules. By leveraging these specific combinations, the full potential of GNNs can be harnessed by professionals in the building and civil infrastructure domain, enhancing operational efficiency, safety, and sustainability of infrastructure assets.

4.6. Extended GNN functionality utilisation

The exploration into GNN based functionalities used within the existing studies reveals a notable gap in leveraging advanced functionalities of GNNs, particularly in areas of link prediction and in the incorporation of edge features into analysis. To address these shortcomings, we propose potential future applications leveraging link prediction capabilities in conjunction with other GNN functionalities across various underexplored OM scenarios as represented in Table 5. We propose scenarios encompassing data enhancement, system monitoring, optimisation, and forecasting.

While current applications in semantic enrichment primarily focus on improving spatial classifications of buildings [88,152], we see significant potential in extending GNN-based problem formulation to building performance-related matrices. Presently, a wide range of simulation-based software tools are used to determine performance matrices such as space acoustics, fire safety ratings, accessibility ratings, energy ratings, and indoor air quality ratings. We propose leveraging GNN-based node classification functions to determine these ratings using space-related features, with training data derived from established simulation scenarios. This approach would utilise multi-hop neighbourhood analysis, considering nodes 1 and 2 hops away from a central node to determine relevant ratings. This method aligns with the need for more efficient alternatives to current methods [85,86]. Furthermore, it has the potential to enhance BIM and IFC models [87] with further information layers, improve stakeholder engagement [95,96], ensure regulatory compliance [92], and contribute to predictive maintenance strategies with enriched models [78]. However, challenges such as model interpretability [88] and scalability [32] must be addressed by leveraging techniques discussed in Section 4. Labelling of BIM models

Table 5
Proposed future GNN applications.

Process	Scenarios	GNN Nodes	GNN Edges	GNN Function
Building performance-related semantic enrichment	Space acoustic rating Fire safety rating Accessibility rating Energy-efficient design rating	Rooms/Spaces	Spatial and functional relationships	Graph/subgraph/node classification
Building systems data imputation	Missing attribute/sensor data imputation	BIM elements, Sensors, MEP components	Spatial, functional and temporal relationships	Node feature prediction
Anomaly detection	Abnormal energy consumption Indoor air quality anomalies	Spaces, MEP components, Sensors	Spatial, functional and temporal relationships	Graph/subgraph/node classification
Forecasting	Fault prediction	Components/subsystems/faults/maintenance history	Functional interactions, failure patterns	Link prediction
System interdependency analysis	MEP system interdependence	Components/subsystems	Interdependencies	Link prediction

for specific simulation aspects is another possible future research direction. Adequate BIM databases should be developed to account for varying standards across regions. The BIM model database approach could be a supplement to current BIM family/object databases operated by particular building-related item vendors themselves.

We propose extending GNN-based data imputation methods, which have shown significant success in civil infrastructure domains such as water distribution systems [34,114,115] and pavement systems [48], to address the current sparsity of such applications in building-based systems. Specifically, we suggest leveraging GNN algorithms to determine missing values in FM-related attributes derived from design data, as well as to impute missing sensor data during the operations stage. This approach could adapt spatial-temporal GAT-based methods [112] and GCN-based metamodels [34] to the building context. By incorporating physical laws into loss functions [114] and utilising spatial correlations [115], these methods could provide robust predictions of building performance parameters. Furthermore, the integration of heterogeneous data sources [112] and the consideration of multi-dimensional targets [48] could offer more comprehensive imputation scenarios. However, careful attention must be paid to scalability issues [114] and the effective integration of diverse data types [112] when applying these methods to complex building systems. This proposal aims to bridge the gap between civil infrastructure and building-based applications of GNN-driven data imputation, potentially revolutionising how missing data is handled in facility management and building operations.

In the domain of anomaly detection, we propose expanding GNN-based approaches to address critical building management challenges, particularly abnormal energy consumption detection and indoor air quality monitoring. Drawing from successful applications in industrial systems [36] and civil infrastructure [60,120], we suggest leveraging spatial-temporal GNN models to capture complex relationships between different building zones and systems over time. These models could integrate physics-based knowledge of energy systems [36] with data-driven approaches, enhancing the accuracy of anomaly detection. For energy consumption, we propose an unsupervised or semi-supervised learning framework [121] that can identify unusual patterns without relying on predefined fault types, adapting to the dynamic nature of building operations. This approach could represent different building systems as nodes in a graph, with edges reflecting their interdependencies [60]. For indoor air quality, we propose extending the semantic enrichment techniques to continuous monitoring, potentially adapting frameworks such as BuiltNet [49] to analyse temporal variations in air quality parameters. However, careful consideration must be given to data preprocessing challenges [49] and the efficient assignment of weights in adjacency matrices to accurately capture the complex interactions within building systems. Moreover, computational efficiency should be a key focus to ensure real-time applicability [121], particularly crucial for the timely detection of energy anomalies and air quality issues.

Several studies that utilised GNN-based functionalities proved the feasibility of GNNs for system monitoring [49,121,129], and further in

fault classification [126] scenarios. We propose enhancing fault prognostics by constructing maintenance-oriented knowledge graphs [164], leveraging multi-source heterogeneous data including sensor signals, images, and maintenance logs. The knowledge graphs would represent heterogeneous data relevant to components, fault symptoms, causes, solutions, and maintenance strategies as nodes, with edges illustrating their relationships, similar to the graph structure used in chiller plant fault classification [124]. A heterogenous GNN model [65] with possible improvements in attention [42] could be designed to predict potential links within the maintenance-oriented knowledge graphs, providing potential fault explanations and maintenance solutions [124]. The model's training could employ a semi-supervised learning approach, using existing edges as positive samples and non-existent edges as negative samples. Performance evaluation using metrics such as Mean Reciprocal Rank (MRR) and Hit@N would assess the model's ability to rank correct links higher [164]. Future research should focus on incorporating temporal aspects. This approach has the potential to significantly enhance predictive maintenance strategies, moving beyond current fault classification methods to proactive fault prediction and mitigation in building and infrastructure management.

Building upon current GNN applications in fault classification, we propose leveraging GNN-based link prediction functions to determine interdependencies between building systems, a crucial aspect of comprehensive facility management. This approach can be implemented across multiple dimensions, initiated by leveraging design-based FM attributes used for preventive maintenance scheduling. By representing building systems as nodes and their relationships as edges, similar to the graph structure used in chiller plant fault classification [124], the interdependencies of preventive maintenance schedules can be modelled. This model can be dynamically enhanced by incorporating performance data from IoT systems as additional features, similar to the integration of multiple data sources in concrete crack detection [128]. Such an approach would enable the determination of dynamic interdependencies based on performance matrices and reactive maintenance needs, similar to the method of adaptive fault diagnosis method for complex energy networks [125]. Furthermore, this concept could be extended to leverage building performance simulations, transforming simulation outputs into graph-based features for analysis, similar to the approach used in structural damage detection with vibration signals [127]. This multi-faceted approach would benefit from multi-task learning techniques, as demonstrated by the Cross-Task GNN in sewer pipe inspection [126], allowing for simultaneous analysis of various interdependency types. However, careful consideration must be given to challenges such as over-smoothing in deep GNN architectures [124], handling imbalanced data across different maintenance scenarios [126], and ensuring model generalisability through comprehensive validation [129]. The proposed methods are human-in-the-loop systems. Engineers need to interpret predictions and validate the accuracy of predictions. GNN-based analysis should be used as a supplementary tools for existing workflows.

4.7. GNNs for building and civil infrastructure digital twins

GNNs are identified as a crucial enabler of digital twins in this review. This synergy is rooted in several key characteristics that align with the demands of digital twin technology. The complex and intertwined nature of relationships within digital twins is mirrored by the inherent strength of GNNs in modelling such intricate networks. GNNs excel at understanding and representing the multifaceted interconnections between different components of a system, which is crucial for digital twins that aim to replicate the complex dynamics of their physical counterparts [143]. The dynamic nature of digital twins, which evolves with the introduction of new data and components, is well-supported by GNNs. These networks could incorporate new nodes and edges, allowing the digital twin to grow and adapt without the need for complete retraining. The heterogeneity present in digital twins, characterised by a diverse array of components and data types, demands a versatile approach to data integration. GNNs meet this demand with their ability to process and integrate multimodal data, enabling the creation of a comprehensive and accurate model that encompasses a variety of data types from simple sensor data to complex operational metrics. Scalability is another critical factor where digital twins could vary greatly in size and complexity. GNNs' efficiency in handling large-scale networks ensures that the computational model of a digital twin could scale accordingly without a loss in performance.

These technical strengths of GNNs further pave the way for their application across different stages of digital twin development as represented in Fig. 12. Starting with the 'Descriptive Twin', GNNs enhance the base-level representation with improved semantic understanding and data completion. As we move to the 'Informative Twin', the dynamic capabilities of GNNs become more pronounced, offering opportunities for real-time analytics and actionable insights [175]. The progression to the 'Predictive Twin' leverages GNNs' ability to forecast future states, enabling proactive maintenance and optimisation strategies. The ultimate goal is the development of 'Comprehensive' and 'Autonomous Twins', where GNNs support the creation of self-learning systems capable of making independent decisions. Although the research in the current domain towards such progression is lacking, vital insights could be acquired from other domains such as manufacturing [176]. Therefore, as highlighted in earlier sections, the improvements in data interoperability, explainability and scalability integrated with supporting technologies as per the proposed implementation workflow (Fig. 9) could be identified as essential sub-categories of research that are necessary for the future, which includes the acquisition of expertise from the domain of computer science.

In concluding Sections 3 and 4, Fig. 13 illustrates the alignment of research gaps, questions, and findings. The study addresses three primary research gaps: (1) underexplored GNN applications in building and civil infrastructure OM, investigated through questions 1 and 2, identifying relevant applications for integrated OM scenarios from initial project phases and operational phases; (2) distinct characteristics of GNN applications in OM, explored via questions 3 and 5, highlighting specific limitations, feature engineering aspects, and data interoperability concerns; and (3) future development scenarios with a focus on

explainability enhancement, examined through questions 4 and 5, focusing on GNN advancements in explainability, network scaling, extended functionalities, and digital twin integration. The figure also outlines key future research directions derived from questions 3 and 4, building upon the identified scenarios and addressing the gaps in current GNN applications within the building and civil infrastructure domain.

5. Conclusion

This systematic review, conducted within the PRISMA framework, presents a comprehensive analysis of the disruptive capabilities of GNNs for OM enhancement in the building and civil infrastructure domain. Addressing the first research question, our study identified the application of GNNs in integrated facilities management approaches for data enhancement, recognising techniques ranging from semantic enrichment of BIM, numerous data imputation scenarios, and semantic segmentation of point clouds aiming to enhance both data quality and completeness. Expanding upon the second research question, our analysis revealed the utilisation of GNN algorithms in operational scenarios to enhance OM practices. A notable focus was found on anomaly detection, fault classification, system optimisation, and forecasting, particularly in building energy-related contexts. The major area of focus in existing studies was given to operational enhancement scenarios.

These applications were further explored to identify the methodological optimisations that made GNN feasible in the current domain and the explainability enhancement scenarios to improve the validity of the outputs for engineering applications aligning with the third and fourth research questions. While basic GNN architecture and hyperparameter optimisation were commonly observed across numerous studies, feature engineering emerged as a central aspect. This highlights the critical significance of carefully selecting and encoding relevant features which is instrumental in distinguishing how the same GNN algorithm performs across various prediction scenarios. Optimising GNN architecture remained pivotal, balancing model complexity with the risk of overfitting. Techniques such as graph dropout, residual connections, and skip connections were identified as possible solutions for overfitting. XAI-based methods were identified to improve the validity of the output for engineering applications, fostering stakeholder trust and aiding in diagnosing model bias. PIGNNs further facilitate research in XAI, integrating physical principles into GNN architecture. KGs and ontologies further enhance explainability by explicitly representing entities and relationships, providing a second layer of validation for GNN predictions.

Among further future research directions adhering to research question 5, scenarios for enhancing data interoperability and scalability, along with explainability enhancement, were identified. Automated graph generation was identified as a key research direction in ensuring the efficiency of GNN-based prediction scenarios. Handling the multitude of heterogeneous data in the building and civil infrastructure domain emphasises the need for further sophisticated GNN models with an emphasis on reducing the computational cost. Real-time data processing remains crucial, especially in structural integrity monitoring and energy demand forecasting. Scalability in data handling has been

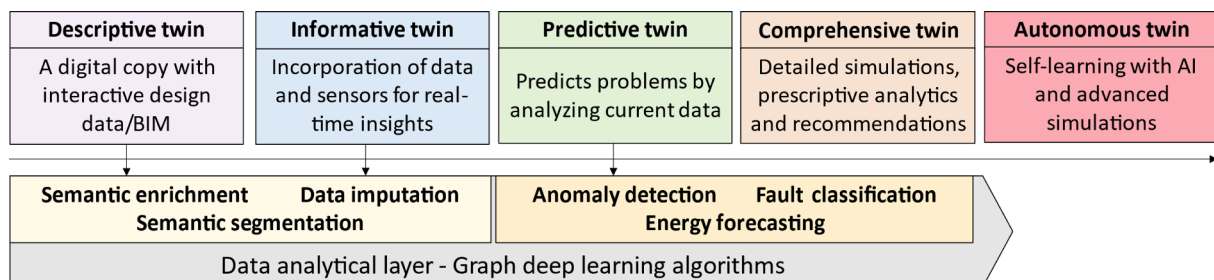


Fig. 12. Digital twin maturity alignment with the identified GNN-based tasks from the current review.

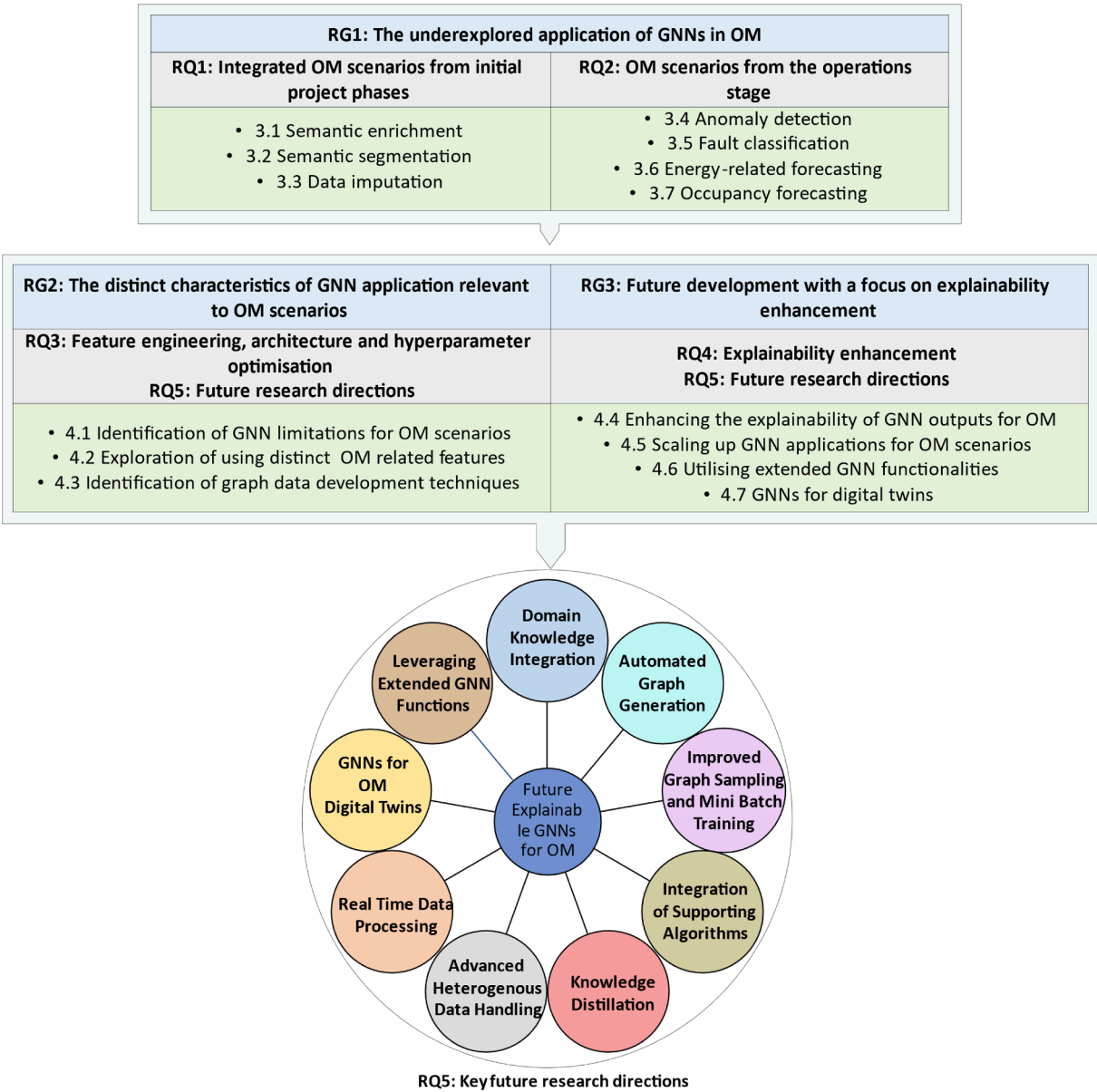


Fig. 13. The alignment of the research gaps, research questions and key findings (RG – Research Gap, RQ – Research Question).

identified as crucial, with techniques such as graph sampling and mini-batch training providing means for implementation across large-scale building and infrastructure projects. Additionally, deploying offline training methods enhances scalability for resource-constrained environments. Supporting algorithms such as LSTMs/GRUs, reinforcement learning, and optimisation algorithms have been identified to overcome GNN limitations in handling temporal and decision-making aspects. Autoencoders and graph transformers have also been identified to complement GNNs by enabling efficient feature extraction and capturing global interactions across complex systems. Development of specific workflows targeting building performance-based semantic enrichment, building systems data imputation, and interdependency prediction are also highlighted as further future research directions. The review further emphasises on the symbiotic relationship between GNN-based analysis and digital twin data analysis, with GNNs’ strengths aligning perfectly with digital twin data analysis demands.

CRediT authorship contribution statement

Sajith Wettewa: Writing – review & editing, Writing – original draft,

Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Lei Hou:** Writing – original draft, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Conceptualization. **Guomin Zhang:** Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix A. Literature search criteria

I. Scopus search criteria

TITLE-ABS-KEY (gnn OR “Graph Neural Network” OR gcn OR “Graph Convolutional Network” OR “Graph Attention Network” OR “GraphSAGE” OR “Graph Convolution Neural Network” OR “Graph Recurrent Neural Network” OR “Graph Auto Encoder” OR “Graph Generative Network” OR “Graph Spatial-Temporal Network” OR “Adaptive Graph Convolutional Neural Network” OR “Graph Long Short-Term Memory Network” OR “Graph LSTM” OR “Gated Graph Neural Network”) AND TITLE-ABS-KEY (“maintenance” OR “buildings” OR “infrastructure” OR “operation and maintenance” OR “civil infrastructure” OR “structural health” OR “digital twin” OR “virtual twin” OR CPS OR “cyber physical system” OR BIM OR “building information model*” OR “construction” OR “design” OR “planning” OR “management”) AND (LIMIT-TO (SUBJAREA, “ENGI”) OR LIMIT-TO (SUBJAREA, “ENVT”) OR LIMIT-TO (SUBJAREA, “ENER”)) AND (LIMIT-TO (DOCTYPE, “ar”) OR LIMIT-TO (DOCTYPE, “cp”) AND (LIMIT-TO (LANGUAGE, “English”)).

II. Web of Science search criteria

Results for TS=(“maintenance” OR “buildings” OR “infrastructure” OR “operation and maintenance” OR “civil infrastructure” OR “structural health” OR “digital twin” OR “virtual twin” OR CPS OR “cyber physical system” OR BIM OR “building information model*” OR “construction” OR “design” OR “planning” OR “management”) AND TS= (gnn OR “Graph Neural Network” OR gcn OR “Graph Convolutional Network” OR “Graph Attention Network” OR “GraphSAGE” OR “Graph Convolution Neural Network” OR “Graph Recurrent Neural Network” OR “Graph Auto Encoder” OR “Graph Generative Network” OR “Graph Spatial-Temporal Network” OR “Adaptive Graph Convolutional Neural Network” OR “Graph Long Short-Term Memory Network” OR “Graph LSTM” OR “Gated Graph Neural Network”) and Article (Document Types) and Article (Document Types) and Article (Document Types) and English (Languages) and Article (Document Types) and English (Languages) and Article or Proceeding Paper (Document Types) and Computer Science Interdisciplinary Applications or Engineering Civil or Transportation Science Technology or Engineering Multidisciplinary or Automation Control Systems or Environmental Sciences or Multidisciplinary Sciences or Engineering Industrial or Engineering Environmental or Green Sustainable Science Technology or Construction Building Technology or Transportation or Environmental Studies (Web of Science Categories) and Article or Proceeding Paper (Document Types) and Article or Proceeding Paper (Document Types) and Engineering Civil or Engineering Multidisciplinary or Transportation Science Technology or Automation Control Systems or Environmental Sciences or Multidisciplinary Sciences or Engineering Industrial or Engineering Environmental or Construction Building Technology or Operations Research Management Science or Transportation (Web of Science Categories).

Appendix B. Summary of studies for semantic enrichment tasks

Note – The column 4 is numbered as follows.

1. Number of layers, 2. Batch size, 3. Learning Rate (LR), 4. Number of epochs/iterations, 5. Activation function, 6. Optimiser, 7. Loss function, 8. Other.

Study	Modules	GNN Node; Features	GNN Edge/Edge features	Optimum architecture and hyperparameters
[33]	GCN, MLP	Spaces; Geometric features, Number of adjacent elements, Shape representation	Accessibility	1. 2 layers, 4. Iterations: 1500, 5. Tanh activation, 6. Adam optimiser
[88]	GAT	Elements or spaces in IFC BIM models; Geometric features, Class labels, window quantity, door opening quantity	Orientation information of the edge, elevation between connected spaces	3. LR 0.001, 4. Epochs: 5000, 6. Adam optimiser, 7. Focal loss
[32]	GraphSAGE	Rooms; Node connection features, Number of door/wall connections, If node is open space, Longest/shortest path features, Number of simple paths	Edge connection type, Door material	1. 4 layers, 2. Batch size: 1, 3. LR: 0.005, 4. Epochs: 200, 6. Adam optimiser, 7. Cross-entropy loss
[90]	GCN, Attention layer, Neural Tensor Network (NTN)	Spaces; Space types, adjacency	Adjacency, wall or door existence in boundaries	1. 3 GCN layers, 2. Batch size: 1, 3. LR: 0.001, 4. Epochs: 20, 7. MSE loss 8. Decay: 0.0005, Dropout: 0.5
[91]	GraphSAGE, differentiable pool (DIFFPOOL) GNN, set2set hierarchical pooling algorithm	Architectural components (e.g., rooms); Area, Function type	Adjacency of components	1. 1–2 layers, 2. Batch size: 20–25, 3. LR: 0.01, 4. Epochs: 10,000–1,000
[89]	Subdivision GNN, hybrid geometry optimisation algorithm	Specific rooms; Position, Type, Degree, Boundary	Connectivity between rooms/corners and their boundary status	1. 8 layers, 3. LR: 0.0001, 4. Epochs: 100 5. Sigmoid activation, 6. Adam optimiser, 8. Decay rate: 0.8

Appendix C. Summary of studies for semantic segmentation tasks

Note – The column 4 is numbered as follows.

1. Learning Rate (LR), 2. Batch Size, 3. Number of Epochs, 4. Other.

Reference	Modules	GNN Node; Features	GNN Edge/Edge features	Optimum architecture and hyperparameters
[100]	Graph Recurrent Attention Networks (GRAN), GRU, GCN	Vertices/Points; Local geometric features around each vertex	Connections based on the BIM model	1. LR:0.002,0.0001, 2. Batch size:1, 3. Epochs:20,000 and 80,000 Other: Momentum:0.9, Adam optimiser, 4. 2-layer GCN
[102]	GCN	Building elements; Feature vector encoding geometric properties/shape features of building element	Topological/ connectivity relationships between elements	
[96]	Local Embedding Super-Point Graph (LE-SPG), Gated Integration Graph Convolutional Network (GIGCN), Gated Hidden States Integration (GHSI), Adaptive Weighted Cross Entropy (AWCE), GRU	Superpoints; Embeddings of superpoints extracted by LSENet	Geometric relationships between neighbouring superpoints	1. LR: 0.001, 2.Batch size: 4, 3. Epochs: 300, 4. Number of GRU layers: 3, GRU hidden state dimension: 64, Number of GRU attention heads: 8, GRU dropout probability: 0.1, α in Adaptive Weighted Cross Entropy loss: 0.7, λ for out-of-distribution loss: 0.1
[145]	BIM-GNN classifier, BIM2Graph module, ifc2vec technique.	Building elements; BIM-based: Element type, floor, Graph-based: Node degree, eigenvector centrality, clustering coefficient, Scan-based: BaselineML features, PointNet features	Relationship types between nodes	1. LR: 0.005, 3. Epochs: 10,000 with early stopping, 4. 2 layers with 16 attention heads each, Dropout probability: 0.2, Hidden layer size: 16, Output layer size: 4, Non-linearity: PReLU, Loss function: Negative log likelihood, Adam optimiser with weight decay 0.0005
[98]	Dynamic graph-based convolutional neural network (DGCNN)	Points; 9D vector for each point	Edges based on Euclidean distance between points	1. LR: 0.001, 2. Batch size: 8, 3. Epochs: 50, 4. Momentum: 0.9, Adam optimiser
[99]	Dynamic graph-based convolutional neural network (DGCNN)	Points; 3D coordinates of each point in the coal mining face	Relative position of neighbouring points	4. Leaky ReLU activation
[97]	Dynamic graph-based convolutional neural network (DGCNN), PointNet, GRU	Superpoints; Features from PointNet architecture	Shape and size comparisons of adjacent superpoints	1. LR: 0.01, 2. Batch size: 4, 3. Epochs: 350, 4. Downsampling voxel size: 0.03 m, Number of nearest neighbours: 10, PointNet layers: MLPs with widths 64,64,128,128,256, Regularisation strength: 0.03, GRU iterations: 10, Adam optimiser
[101]	SPG, GRU	Superpoint clusters	Length ratio, surface ratio, volume ratio, point count ratio	3. Epochs:200, 4. PointNet: 128-dim encoding, 32-dim decoding, Graph RNN: 128 points per superpoint, 512 superpoints for graph RNN

Appendix D. Summary of studies for data imputation tasks

Note – The column 4 is numbered as follows.

1. Batch Size, 2.Learning Rate (LR), 3.Number of Layers, 4. Number of Hidden Units/Layers, 5. Other.

Reference	Modules	GNN Node; Features	GNN Edge/Edge features	Optimum architecture and hyperparameters
[35]	Spectral models: GCN and ChebNet, Two spatial models: GraphSAGE and TAGCN – comparison	Junctions, manholes in wastewater; Diameter, material, Strahler's number, other attributes (if available)	Represent connections	3. Convolution layers: 2, 4. Hidden units: 20 in layer 1; classes in layer 2, Other: Activation: ReLU and LogSoftmax, Filter size for ChebNet: 10–40
[113]	Spatio-Temporal Graph Convolutional Network (T-GCN), Graph Signal Processing (GSP), LSTM, GCN	Sensors; Time series sensor data (strain, vibration)	Correlation or DTW distance between time series data	3. GCN Layers: 2 with 8 filters each, 4. LSTM Layers: 2 with 50 filters each, Other: Dense Output Layer: 42 units
[48]	GraphSAGE	Pavement sections; Pavement condition indicators, time since last treatment, traffic, road work records, climate zones, pavement type	Connections based on reference markers	3. Layers: 2, 4. Hidden channels per layer: 256
[47]	GNN, GAT	Pipes; Age, length, diameter, material, coating material, soil properties, tree canopy coverage, elevation/slope, degree, PageRank, clustering coefficient	Physical connections and geographical proximity	1. Batch size: 3000–2000, 2. LR: 0.001–0.0001, 3. GNN layers: 3 to 4, Other: ρ in graph construction: 100 m to 45 m, k in multi-hop aggregation: 2, Dropout rate: 0.2–0.1, Optimiser: Adam
[112]	GAT, GCN	Sensor (spatial), Time step (temporal); Monitoring data at each time step	Relationships between sensors or time steps	Other: Attention heads: 4, Spatiotemporal attention blocks: 2, Activation functions: Tanh and ELU, Loss function: RMSE, Training methodology: k-fold cross validation, Attention mechanisms compared: Dot product, concatenation, addition, Optimiser not specified
[115]	GCN (SCGCN-ONE, SCGCN-PCC, SCGCN-AC, SCGCN-HCC, SCGCN-HAC), Self-attention mechanism	Sensors; Water demand time series data	Spatial correlation coefficients (PCC, AC, HCC, HAC), Flow resistance between sensors	2. LR: 0.001, 1. Batch size: 16–128, 4. Neurons in hidden layers: 64–128, Other: Dimensionality of aggregated data: 16–128, Attention heads: 10–14, Models:
[114]	GNN	Junctions and reservoirs; Junction indicator, demand, measurement indicator, measured head value	Pipes – start and end nodes of pipe, loss coefficient	Other: Latent dimension: 20, Correction update layers: 20, Hidden layers in MLP: 1, Discount factor: 0.9, Update coefficient: 0.01

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(continued)

Reference	Modules	GNN Node; Features	GNN Edge/Edge features	Optimum architecture and hyperparameters
[34]	GCN	Junctions or points in the water network; Pressure values for NG, flow rate values for LG	Connectivity between nodes/pipes	1. Batch Size: 32, 3. GCN Architecture: 3 layers (32, 64, 32 units)

Appendix E. Summary of studies for anomaly detection tasks

Note – The column 4 is numbered as follows.

1. Batch size, 2. Learning Rate (LR), 3. Number of epochs 4. Other.

Reference	Modules	GNN Node; Features	GNN Edge/Edge features	Optimum architecture and hyperparameters
[121]	1DCNN, GraphTransformer, Contrastive learning	Sensors; Raw time-series vibration data	Adjacency matrix based on sensor connectivity	1. Batch size: 256, 2. Learning rate: 0.001, Other: Graph Transformer layers: 2, Attention heads: 4, Output size: 128, Temperature coefficient: 0.3, 1D CNN kernels: 32, 1D CNN kernel size: 100, Classifier hidden layer size: 256, Optimizer: Adam
[49]	GCN, LSTM	Regions in thermal images; Encoded temperature values from thermal images	Similarity and spatial/temporal distance between image regions	3. Epochs: Until convergence, Other: GNN layers: 2, Number of clusters: 3
[60]	GCN, 1DCNN	Cable dynamometers; Time series of cable force measurements	Learnable adjacency matrices	Other: Features per node (F): 1, Nodes (N): 42, Activation functions: Tanh and ReLU, Loss function: MSE + penalty terms
[120]	RNN, GCN, Variational autoencoder	Sensors; Sensor time series data	Distance-based connectivity in the adjacency matrix	1. Batch size: 30, 3. Epochs: 100, Other: Smoothness regularisation (α): 0.6, Similarity regularisation (β): 0.02, Graph convolution filters (K): 3, Dropout rate: 0.2
[36]	GRU, GCN, graph attention	Process variables like sensors and actuators in ICPSs; Embeddings from time series data	Structural dependencies between pairs of variables	1. Batch size: 64, 2. LR: 0.001/0.01, 3. Maximum epochs: 50, Other: Embedding dimension (d_emb): 32/64, Hidden state dimension (d_h): 128, Regularisation weights (λ_1, λ_2): Varied, Window size (w): 15, 1D convolution kernel size (k): 15

Appendix F. Summary of studies for fault classification tasks

Note – The column 4 is numbered as follows.

1. Number of Layers, 2. Learning Rate (LR), 3. Number of Epochs, 4. Dropout Rate, 5. Other.

Reference	Modules	GNN Node; Features	GNN Edge/Edge features	Optimum architecture and hyperparameters
[124]	GCN, FCNN, Cheb, SAGE, GAT, GCN2 comparison	Data samples; Operating conditions (temperatures, flowrates, setpoints). Each node represents one data sample.	Edge weights representing similarities between data samples, calculated using Euclidean distances.	1. CGN layers: 1–4, 2. Initial LR: 0.1, 4. Dropout rate: 0.2, Other: k value for kNN graph generation: 5 or 10, LR scheduler: Reduce on Plateau, Imbalanced class weights: Reciprocals of class frequencies
[126]	GCN, GAT	Classes across tasks (e.g., defect types); Class embeddings for each class in each task.	Adjacency matrix capturing relationships between classes in different tasks, based on co-occurrence stats.	3. Epochs: 100, 2. LR: 0.01, 5. Other: Threshold σ : 0.4, Activation: LeakyReLU (–0.2), Optimiser: SGD, Momentum: 0.9, Weight decay: 0.001, LR decay: Every 40 epochs by 10
[128]	GCN, ResNet-101, Hybrid residual network	Crack feature maps extracted from images using ResNet-101.	Adjacency matrix encoding co-occurrence relationships between labels of crack images.	3. Epochs: 100, 2. Initial LR: 0.01, Other: Threshold σ : 0.4, Activation: LeakyReLU (–0.2), Optimiser: SGD, Momentum: 0.9, Weight decay: 0.001, LR decay: Every 40 epochs by 10
[129]	GCN	Feature vectors; Position and normal vector of feature points.	Attention coefficients between different feature points, calculated based on feature similarity.	
[127]	Continuous Wavelet Transform (CWT), Laplacian matrix of the spatial-temporal graph, DGCN, GCN – spectral	Multisensor spatial-temporal graph-based feature vectors; Eigenvalues of the Laplacian matrix derived from spatial-temporal graph constructed from CWT coefficients.	Edges connect nodes with similar CWT coefficients over time, weighted by difference in CWT coefficients.	1. Number of GCN layers (D): 2, 2. Initial LR: 0.0001, 3. Epochs: 500 (for Qatar dataset), Other: Number of output features per layer (N): 64 (1st layer), 30 (2nd layer), Kernel size (K): 3 or 1, Nonlinear activation function: ReLU
[125]	GAT, GRU, GRNN	Key sensor locations of the heating network; Node pressure values and pipeline flow rates.	Edges represent physical connections in pipeline topology, encoded in binary adjacency matrix.	Other: Number of heads in GAT module: 2, Hidden size in GAT module: 6, Number of GRU gated units: 24, Hidden size in GRU: 4, Activation functions: LeakyReLU (GAT), Sigmoid (GRU)

Appendix G. Summary of studies for forecasting tasks

Note – The column 4 is numbered as follows.

1. Layers, 2. Epochs, 3. Learning Rate (LR), 4. Hidden Size/Units/Dimension, 5. Attention Heads, 6. Optimiser, 7. Loss function (including MAE, Cross-entropy, MSE), 8. Kernel Size, and 9. Other.

Reference	Modules	GNN Node; Features	GNN Edge/Edge features	Optimum architecture and hyperparameters
[142]	GNN, Build2Vec	Spaces; Spatial data, Occupancy data, Environmental data	Spatial proximity, Area of influence	9. Other: Random walk length: 50, Walks per node: 50, Context size: 30
[134]	GCN, GatedGCN, MoNet, GAT, GraphSage	Rooms; Room type	Adjacency between rooms	1. Layers: 4–16, 2. Epochs: ≤ 1000 , 3. LR: 0.001, 4. Units: 32–128, 6. Adam optimiser, 7. Loss Function (MAE)
[135]	ST-GCN	Buildings; Building characteristics, Weather data, Energy consumption	Angle, Distance, Shadow orientation between buildings	9. Standard or default GNN values
[37]	GRU, GCN	Heat meters; Heat load time series data	Heat load pattern similarity	4. Hidden Size: 24–26, 8. Kernel Size: 3–9, 9. Other: CNN channels: 23 to 26, Correlated meters: 3 to 20
[46]	Point-GNN- for detecting objects given a point cloud, MLP	Point cloud points; Point coordinates, Feature vectors	Distance-based connectivity	Other: Cross entropy loss, Huber loss
[143]	GraphSAGE, LSTM	Spatial objects, environmental variables, virtual nodes; Sidewalk traffics, Built environment, Environmental variables, Virtual nodes	Connectivity defined by adjacency matrix	2. Epochs: 100, 3. LR: 0.0001, 7. Loss Function: Cross-entropy, 9. Batch size: 128
[139]	GAT, GRU	Clusters of heat load observations; Heat load, Meteorological observations	Attention scores from node features	4. Hidden Size: 24–26, 5. Attention Heads: 2–4, 9. Other: Clusters: 3
[136]	DL-GCN, LSTM	Zones; Zone-based data (temperature, humidity, etc.)	Spatial layout connectivity	1. Layers: 3, 9. Other: Embedding dimension: 35, Chebyshev terms: 1
[144]	AnyLogic software, Hybrid GNN and LSTM framework	Rooms; Room occupancy data	Connectivity between rooms	1. Layers: 5, 2. Epochs: ≤ 200 , 3. LR: 0.001, 9. Other: Neurons per layer: 64
[50]	Temporal trend-aware graph attention network (TTGAT), gated temporal convolution layer (GTC)	Buildings; Historical cooling load data	Spatial correlations between buildings	1. Layers: 4, 2. Epochs: 200, 3. LR: 0.001, 5. Attention Heads: 8, 9. Other: Convolution kernels: 3x1 & 4x1, FC layers: 64
[138]	GNN, RNN	Physical entities (like VAV boxes and sensors); Operational data of physical entities, IoT sensor data	Physical connections and spatial relationships	1. Layer: GCN layer, DC layer, graph attention layer/ GAT layer, recurrent layer (LSTM), fully connected layer, 5. Attention Heads: 10 for GAT 9. Loss function RMSE, Recurrent units in each LSTM: 15, Activation: Sigmoid

Appendix H. Limitations for GNN applications in the semantic enrichment domain

Limitation	Description	Reference
Data interoperability		
Manual annotation	Required due to lack of standardised formats across datasets.	[89]
Dataset construction process	Bypasses direct operation on BIM files, building graphs manually from 2D/3D plans instead.	[32]
Scalability		
Slower recommendation system speeds	Complexity of GNN models leads to slower recommendation system speeds as dataset sizes increase, negatively impacting user experience.	[90]
Ineffective scaling for larger datasets	SAGE-E model struggles to scale effectively for larger datasets, a common concern across many GNN models.	[32]
Need for additional computational resources and optimisation	Necessary for handling larger datasets.	[90]
Explainability		
Opaque clustering mechanisms in DIFFPOOL model	Clusters neighbouring nodes in an opaque manner, leaving clustering mechanisms unexplained.	[91]
Variability in prediction performance across space function classes	Performance varies without clear explanations.	[88]
Other limitations		
Class imbalance	Significantly impacts prediction performance.	[88]
Need for better feature integration	Necessary to improve model performance.	[88]
Generalisability issues	Requires additional research to ensure effectiveness.	[91]
Theoretical limitations of generating certain room layouts	Unable to generate certain room layouts due to theoretical constraints.	[89]
Specific case failures due to complex boundaries	Failures observed in specific cases.	[89]
Difficulties in determining spatial design features from model outputs	Challenging to identify which spatial design features can be effectively obtained and utilised from model outputs.	[91]

Appendix I. Limitations for GNN applications in the semantic segmentation domain

Limitation	Description	Reference
Data interoperability		
Extracting local features	Ineffective extraction of local features in CMF point cloud.	[99]
Lack of bridge information models	Difficulty in generating synthetic bridge point clouds due to the absence of BRIMs.	[97]
Scalability		
Node-wise similarity computation	High computational complexity for larger datasets due to quadratic node-wise similarity computation.	[101]
Computational complexity	High computational load for processing large-scale 3D point cloud data.	[96]
Multi-scale and long-term geometric contexts	Inability to handle multi-scale and long-term geometric contexts in pavement point clouds.	[104]
Need for improved robustness	Requirement for enhanced robustness and performance in diverse environments.	[96]
Explainability		
Boundary discrimination challenges	Difficulty in consistently distinguishing boundaries between adjacent components.	[97]
Other limitations		
Sampling cube size determination	Challenge in selecting optimal size for sampling cubes, affecting detection accuracy and computational load.	[146]
Feature encoding for complex structures	Difficulty encoding features for cracks with complex structures and low connectivity.	[105]
Default hyperparameters	Potential suboptimal model performance due to reliance on default hyperparameters.	[105]
Synthetic datasets performance limitation	Limited improvement in performance due to geometry and colour differences in synthetic datasets.	[103]

Appendix J. Limitations for GNN applications in the data imputation domain

Limitation	Description	Reference
Data interoperability		
High correlation of heterogeneous data	Heterogeneous data did not improve performance due to sufficient information in homogeneous data.	[112]
Limited scope of data	Focused only on temperature data, requiring a global perspective and better data fusion methods.	[112]
SCGCN-PCC model performance	Pearson correlation coefficient's limitations highlight the need for improved quantification methods.	[115]
Data availability and privacy	Limited access to high-quality data from water companies due to privacy and regulatory issues.	[115]
Scalability		
Fixed-size neighbourhood limitation	SAGEConv's fixed neighbourhood size limits performance compared to ChebConv.	[35]
Scalability to larger networks	More tests needed on larger networks with complex components like valves and pumps.	[114]
Explainability		
Uncertainty in data-driven models	High uncertainty in estimation models not physically based, causing practical issues in water supply management.	[34]
Need for adaptive data recovery techniques	Essential to develop techniques for addressing missing data in random sensors.	[112]
Other limitations		
Recognition of spatial relationships	Recognises only linear sections, suggesting inclusion of latitude/longitude data.	[48]
Interdependencies between condition indicators	Does not address interdependencies; combining indicators could improve performance.	[48]
Convergence issues	Slow and inconsistent convergence with current loss function, indicating need for alternative metrics.	[114]
Homogeneous sensors	Limited to homogeneous sensors; diverse data integration needed.	[114]
Performance on rapid time-varying vibration response	Needs verification for rapid time-varying vibration response.	[112]

Appendix K. Limitations for GNN applications in the anomaly detection domain

Limitation	Description	Reference
Data Interoperability		
Adjacency matrix limitations	Only considers sensor distances, and ignores mechanical relationships.	[120]
Low-cost thermal cameras	Compromised temperature measurement accuracy.	[49]
Noise in preprocessing	Inconsistent noise removal, affecting thermal data.	[49]
Incomplete thermal interaction capture	Inadequate capture of complex thermal interactions.	[49]
Scalability		
High computational costs	High costs and low performance for time series modelling.	[119]
High time complexity	Significant CPU time required, especially in semi-SDD models.	[121]
Challenges in real-time application	Lengthy inference times, needs substantial computational resources.	[121]
Need for dynamic adaptation	Models must adapt to time-varying data.	[36]
Long training times	Some models, like semi-SDD, have significantly longer training times.	[121]
Explainability		
Reduced effectiveness in extracting spatial features	Difficult to extract spatial anomalies in abnormal time series.	[120]
Lack of sensor interaction handling	Sensor interactions not fully leveraged, impacting performance.	[121]
Underperformance without prior knowledge integration	Models perform worse without prior knowledge.	[36]
Limited complex interaction capture	Weight assignments in matrices miss complex interactions, limiting explainability.	[49]
Other Limitations		
Overlooked temporal correlations	Temporal correlations often ignored, requiring LSTM.	[119]
Dependence on historical data	Performance drops with invalid historical data.	[120]
Inability to localise anomalies in advance	Unable to predict and localise anomalies early.	[119]
Propagation of anomalies	Requires variable causality for effective anomaly localisation.	[36]

Appendix L. Limitations for GNN applications in the fault classification domain

Limitation	Description	Reference
Data interoperability		
Data quality and missing data	Model sensitivity increases with 50 %-80 % missing temporal samples.	[125]
Complex data preprocessing	Time-frequency analysis and matrix decomposition are time-consuming.	[127 128]
Dependence on large and balanced datasets for training.	High imbalance ratios and limited training data negatively affect accuracies.	[124]
Limited training data	High imbalance ratios and limited training data negatively affect accuracies.	[126]
Explainability		
Importance of model interpretability.	LightGBM surrogate models are used for predictive modelling and interpretation.	[124]
Sensitivity to noise and missing data affects explainability.	Explainability is reduced due to noise sensitivity and missing data.	[128]
Other		
Challenges in applying models to real-world scenarios.	Difficulties in identifying unknown damage types and locations.	[127]
Class imbalance	Poor performance on imbalanced classes and differences between validation and test splits.	[126]
Generalisability issues	Performance discrepancies between training and validation sets indicate poor generalisation.	[129]

Appendix M. Limitations for GNN applications in the forecasting domain

Limitation	Description	Reference
Data interoperability		
Manual graph design	Time-consuming and labour-intensive for large networks.	[50]
Manual data processing	Requires automated object detection for graph conversion.	[134]
Data quality dependency	Performance relies on high-quality, abundant data.	[51]
Training data dependency	Requires large amounts of training data.	[139]
Scalability		
Computational efficiency	High computational resources needed for large networks.	[139]
Training and update resources	Significant cost for training and updating models.	[37]
Propagation cost	High cost for multi-layer information propagation.	[143]
Fixed graph structure	Does not capture dynamic topology changes.	[137]
Model complexity	High complexity adds noise and complicates results.	[144]
Explainability		
Black box models	Difficult to understand and trust predictions.	[139]
Lack of interpretable explanations	Predictions are not easily interpretable.	[37]
Complexity of structure	Hard to explain complex GNN structures.	[143]
Practical application trust	Interpretability issues limit practical use and trust.	[138]
Limited evaluation scope	Needs broader evaluation across different areas.	[135]
Other limitations		
Real-life scenario deployment	Practical challenges in device provision and privacy.	[142]
Generalisability issues	Models may overfit to specific training data.	[138]
Granularity and overfitting	Issues with class assignment and overfitting due to repetitive geometries.	[46]

References

- [1] M. Gori, G. Monfardini, F. Scarselli, A new model for learning in graph domains, in: Proceedings. 2005 IEEE International Joint Conference on Neural Networks, 2005., 2005: pp. doi: doi: 729–734 vol. 2. doi: 10.1109/IJCNN.2005.1555942.
- [2] J. Zhou, G. Cui, S. Hu, Z. Zhang, C. Yang, Z. Liu, L. Wang, C. Li, M. Sun, Graph neural networks: A review of methods and applications, *AI Open* 1 (2020) 57–81, <https://doi.org/10.1016/j.aiopen.2021.01.001>.
- [3] M.M. Bronstein, J. Bruna, Y. LeCun, A. Szlam, P. Vandergheynst, Geometric deep learning: Going beyond Euclidean data, *IEEE Signal Process Mag.* 34 (2017) 18–42, <https://doi.org/10.1109/MSP.2017.2693418>.
- [4] A. Derrow-Pinion, J. She, D. Wong, O. Lange, T. Hester, L. Perez, M. Nunkesser, S. Lee, X. Guo, B. Wiltshire, Eta prediction with graph neural networks in google maps, in: Proceedings of the 30th ACM International Conference on Information & Knowledge Management, 2021: pp. 3767–3776. <https://doi-org.ezproxy.lib.rmit.edu.au/10.1145/3459637.3481916>.
- [5] R. Ying, R. He, K. Chen, P. Eksombatchai, W.L. Hamilton, J. Leskovec, Graph convolutional neural networks for web-scale recommender systems, in: Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, 2018: pp. 974–983. <https://doi-org.ezproxy.lib.rmit.edu.au/10.1145/3219819.3219890>.
- [6] M. Zitnik, M. Agrawal, J. Leskovec, Modeling polypharmacy side effects with graph convolutional networks, *Bioinformatics* 34 (2018) 1457–1466, <https://doi.org/10.1093/bioinformatics/bty294>.
- [7] J. Gilmer, S.S. Schoenholz, P.F. Riley, O. Vinyals, G.E. Dahl, Neural message passing for Quantum chemistry, in: Proceedings of the 34th International Conference on Machine Learning - Volume 70, JMLR.org, Sydney, NSW, Australia, 2017: pp. 1263–1272. doi: 10.48550/arXiv.1704.01212.
- [8] Brian. Atkin, Adrian. Brooks, Total facilities management, in: Total Facilities Management, 2nd ed., Blackwell Pub., Malden, MA, 2005.
- [9] Barrie. Chanter, Peter. Swallow, Building maintenance management, in: Building Maintenance Management, Blackwell Science, Oxford., 1996.
- [10] M. Marzouk, M. Zaher, Artificial intelligence exploitation in facility management using deep learning, *Constr. Innov.* 20 (2020) 609–624, <https://doi.org/10.1108/CI-12-2019-0138>.
- [11] H. Farzaneh, L. Malehmirchegini, A. Bejan, T. Afolabi, A. Mulumba, P.P. Daka, Artificial intelligence evolution in smart buildings for energy efficiency, *Appl. Sci.* 11 (2021) 763, <https://doi.org/10.3390/app11020763>.
- [12] C.H. Wong, M.H.A. Samad, N. Taib, Potential and limitation of AI system in building services and control management system, *IOP Conf. Ser.: Earth Environ. Sci.* 881 (2021) 012044, <https://doi.org/10.1088/1755-1315/881/1/012044>.
- [13] M.S. Orooj, M.M. Latifi, A review of embedding artificial intelligence in internet of things and building information modelling for healthcare facility maintenance management, *Energy Environ. Res.* 11 (2021) 31, <https://doi.org/10.5539/eer.v11n2p31>.
- [14] I. Osadcha, A. Jurelionis, P. Fokaidas, Geometric parameter updating in digital twin of built assets: A systematic literature review, *J. Build. Eng.* 73 (2023) 106704, <https://doi.org/10.1016/j.jobe.2023.106704>.
- [15] H. Chen, L. Hou, G. (Kevin) Zhang, S. Moon, Development of BIM, IoT and AR/VR technologies for fire safety and upskilling, *Automation in Construction* 125 (2021) 103631. doi: 10.1016/j.autcon.2021.103631.
- [16] M. Mahmoodian, F. Shahrivar, S. Setunge, S. Mazaheri, Development of digital twin for intelligent maintenance of civil infrastructure, *Sustainability* 14 (2022) 8664, <https://doi.org/10.3390/su14148664>.
- [17] J. Zhao, H. Feng, Q. Chen, B. Garcia de Soto, Developing a conceptual framework for the application of digital twin technologies to revamp building operation and maintenance processes, *J. Build. Eng.* 49 (2022) 104028, <https://doi.org/10.1016/j.jobe.2022.104028>.
- [18] W. Yao, D. Li, L. Gao, Fault detection and diagnosis using tree-based ensemble learning methods and multivariate control charts for centrifugal chillers, *J. Build. Eng.* 51 (2022) 104243, <https://doi.org/10.1016/j.jobe.2022.104243>.
- [19] X. Zhou, T. Yang, L. Liang, X. Zi, J. Yan, D. Pan, Anomaly detection method of daily energy consumption patterns for central air conditioning systems, *J. Build. Eng.* 38 (2021) 102179, <https://doi.org/10.1016/j.jobe.2021.102179>.

- [20] V. Giuchene Lu, A.K. Parlikad, P. Woodall, G.D. Ranasinghe, J. Heaton, Developing a Dynamic Digital Twin at a Building Level: using Cambridge Campus as Case Study, in: International Conference on Smart Infrastructure and Construction 2019 (ICSIC), ICE Publishing, 2019: pp. 67–75. doi: 10.1680/icsic.64669.067.
- [21] A. Holzinger, B. Malle, A. Saranti, B. Pfeifer, Towards multi-modal causability with Graph Neural Networks enabling information fusion for explainable AI, *Information Fusion* 71 (2021) 28–37, <https://doi.org/10.1016/j.inffus.2021.01.008>.
- [22] Y. Ektefaie, G. Dasoulas, A. Noori, M. Farhat, M. Zitnik, Multimodal learning with graphs, *Nat. Mach. Intell.* 5 (2023) 340–350, <https://doi.org/10.1038/s42256-023-00624-6>.
- [23] Y. Xu, Y. Zhou, P. Sekula, L. Ding, Machine learning in construction: From shallow to deep learning, *Develop. Built Environ.* 6 (2021) 100045, <https://doi.org/10.1016/j.dibe.2021.100045>.
- [24] L. Hou, H. Chen, G. (Kevin) Zhang, X. Wang, Deep Learning-Based Applications for Safety Management in the AEC Industry: A Review, *Applied Sciences* 11 (2021) 821. doi: 10.3390/app11020821.
- [25] Y. Jia, J. Wang, W. Shou, M.R. Hosseini, Y. Bai, Graph neural networks for construction applications, *Autom. Constr.* 154 (2023) 104984, <https://doi.org/10.1016/j.autcon.2023.104984>.
- [26] Z. Chen, J. Xu, C. Alippi, S.X. Ding, Y. Shardt, T. Peng, C. Yang, Graph neural network-based fault diagnosis: a review, (2021). <http://arxiv.org/abs/2111.08185> (accessed June 19, 2023).
- [27] N.A. Asif, Y. Sarker, R.K. Chakraborty, M.J. Ryan, M.H. Ahamed, D.K. Saha, F. R. Badal, S.K. Das, M.F. Ali, S.I. Moyeen, M.R. Islam, Z. Tasneem, Graph Neural Network: A comprehensive review on non-Euclidean space, *IEEE Access* 9 (2021) 60588–60606, <https://doi.org/10.1109/ACCESS.2021.3071274>.
- [28] J. Kim, S. Chi, Graph neural network-based propagation effects modeling for detecting visual relationships among construction resources, *Autom. Constr.* 141 (2022) 104443, <https://doi.org/10.1016/j.autcon.2022.104443>.
- [29] Y. Fei, W. Liao, X. Lu, H. Guan, Knowledge-enhanced graph neural networks for construction material quantity estimation of reinforced concrete buildings, *Comput. Aided Civ. Inf. Eng.* 39 (2024) 518–538, <https://doi.org/10.1111/mice.13094>.
- [30] R. Chudley, R. Greeno, *Building construction handbook*, Routledge, 2006.
- [31] K. Roper, R. Payant, *The Facility Management Handbook*, AMACOM, 2014.
- [32] Z. Wang, R. Sacks, T. Yeung, Exploring graph neural networks for semantic enrichment: Room type classification, *Autom. Constr.* 134 (2022) 104039, <https://doi.org/10.1016/j.autcon.2021.104039>.
- [33] A. Buruzs, M. Sipetić, B. Blank-Landeshammer, G. Zucker, IFC BIM model enrichment with space function information using Graph Neural Networks, *Energies* 15 (2022), <https://doi.org/10.3390/en15082937>.
- [34] A. Zanfei, A. Menapace, B.M. Brentan, R. Sitzenfrie, M. Herrera, Shall we always use hydraulic models? A Graph Neural Network metamodel for water system calibration and uncertainty assessment, *Water Res.* 242 (2023) 120264, <https://doi.org/10.1016/j.watres.2023.120264>.
- [35] Y. Belghaddar, N. Chahinian, A. Seriai, A. Begdouri, R. Abdou, C. Delenne, Graph convolutional networks: Application to database completion of wastewater networks, *Water* 13 (2021) 1681, <https://doi.org/10.3390/w13121681>.
- [36] W. Wu, C. Song, J. Zhao, Z. Xu, Physics-informed gated recurrent graph attention unit network for anomaly detection in industrial cyber-physical systems, *Inf. Sci.* 629 (2023) 618–633, <https://doi.org/10.1016/j.ins.2023.01.136>.
- [37] Y. Huang, Y. Zhao, Z. Wang, X. Liu, H. Liu, Y. Fu, Explainable district heat load forecasting with active deep learning, *Appl. Energy* 350 (2023), <https://doi.org/10.1016/j.apenergy.2023.121753>.
- [38] R. Jang, W. Collinge, Improving BIM asset and facilities management processes: A Mechanical and Electrical (M&E) contractor perspective, *J. Build. Eng.* 32 (2020) 101540.
- [39] T. Halim, K. Muthusamy, S. Yong Chia, S. Wei Lam, A systems approach in the evaluation and comparison of engineering services applied in facilities management, *Facilities* 29 (2011) 114–132.
- [40] J. Beall, Z. Cherenet, L. Cirolia, N.F. da Cruz, Understanding infrastructure interfaces: Common ground for interdisciplinary urban research? *J. Br. Acad.* (2019) 11–43.
- [41] P. Bocchini, D.M. Frangopol, T. Ummenhofer, T. Zinke, Resilience and sustainability of civil infrastructure: Toward a unified approach, *J. Infrastruct. Syst.* 20 (2014) 04014004.
- [42] T. Hartmann, A. Trappey, Advanced Engineering Informatics - Philosophical and methodological foundations with examples from civil and construction engineering, *Develop. Built Environ.* 4 (2020) 100020, <https://doi.org/10.1016/j.dibe.2020.100020>.
- [43] S.O. Abioye, L.O. Oyedele, L. Akanbi, A. Ajayi, J.M.D. Delgado, M. Bilal, O. O. Akinade, A. Ahmed, Artificial intelligence in the construction industry: A review of present status, opportunities and future challenges, *J. Build. Eng.* 44 (2021) 103299.
- [44] P. Geyer, M.M. Singh, X. Chen, Explainable AI for engineering design: A unified approach of systems engineering and component-based deep learning, *arXiv Preprint arXiv:2108.13836* (2021).
- [45] M.A. Chao, C. Kulkarni, K. Goebel, O. Fink, Fusing physics-based and deep learning models for prognostics, *Reliab. Eng. Syst. Saf.* 217 (2022) 107961.
- [46] R.B. Joshi, A.P. Behera, S. Mishra, eBIM-GNN Fast and Scalable energy analysis through BIMs and Graph Neural Networks, in: 2022 IEEE 7th International Energy Conference (ENERGYCON), 2022: pp. 1–8. doi: 10.1109/ENERGYCON53164.2022.9830484.
- [47] S. Liang, Z. Li, B. Liang, Y. Ding, Y. Wang, F. Chen, Failure Prediction for Large-scale Water Pipe Networks Using GNN and Temporal Failure Series, in: Proceedings of the 30th ACM International Conference on Information & Knowledge Management, ACM, Virtual Event Queensland Australia, 2021: pp. 3955–3964. doi: 10.1145/3459637.3481918.
- [48] L. Gao, K. Yu, P. Lu, Considering the spatial structure of the road network in pavement deterioration modeling, *Transp. Res. Rec.* (2023) 03611981231188373, <https://doi.org/10.1177/03611981231188373>.
- [49] N. Khan, N. Roy, BuiltNet: Graph based Spatio-Temporal Indoor Thermal Variation Detection, in: 2021 20th IEEE International Conference on Machine Learning and Applications (ICMLA), 2021: pp. 1696–1703. doi: 10.1109/ICMLA52953.2021.00270.
- [50] M. Yu, D. Niu, J. Zhao, M. Li, L. Sun, X. Yu, Building cooling load forecasting of IES considering spatiotemporal coupling based on hybrid deep learning model, *Appl. Energy* 349 (2023) 121547, <https://doi.org/10.1016/j.apenergy.2023.121547>.
- [51] Z. Zhao, Z. Chen, Q. Deng, P. Tang, T. Peng, Cooling load prediction based on correlated temporal graph convolutional network, in: 2022 4th International Conference on Industrial Artificial Intelligence (IAI), 2022: pp. 1–6. doi: 10.1109/IAI55780.2022.9976497.
- [52] Y. Gao, H. Yang, P. Zhang, C. Zhou, Y. Hu, Graph neural architecture search, in: International Joint Conference on Artificial Intelligence, 2021.
- [53] S. Jegelka, Theory of graph neural networks: Representation and learning, in: 2022.
- [54] B. Atkin, A. Brooks, *Total Facility Management*, John Wiley & Sons, 2021.
- [55] D.N. Bristow, A.H. Hay, Graph model for probabilistic resilience and recovery planning of multi-infrastructure systems, *J. Infrastruct. Syst.* 23 (2017) 04016039.
- [56] N. Wang, G. Zhu, L. Zhang, P. Shen, H. Li, C. Hua, Spatio-Temporal Interaction Graph Parsing Networks for Human-Object Interaction Recognition, (2021). doi: 10.48550/arXiv.2108.08633.
- [57] N. Dehmamy, A.-L. Barabási, R. Yu, Understanding the representation power of graph neural networks in learning graph topology, *Adv. Neural Inf. Process. Syst.* 32 (2019).
- [58] J.W. Ma, T. Czerniawski, F. Leite, Semantic segmentation of point clouds of building interiors with deep learning: Augmenting training datasets with synthetic BIM-based point clouds, *Autom. Constr.* 113 (2020) 103144, <https://doi.org/10.1016/j.autcon.2020.103144>.
- [59] K. Sun, T. Hong, J. Kim, B. Hooper, Application and evaluation of a pattern-based building energy model calibration method using public building datasets, *Build. Simul.* 15 (2022) 1385–1400, <https://doi.org/10.1007/s12273-022-0891-2>.
- [60] S. Li, J. Niu, Z. Li, Novelty detection of cable-stayed bridges based on cable force correlation exploration using spatiotemporal graph convolutional networks, *Struct. Health Monit.* 20 (2021) 2216–2228, <https://doi.org/10.1177/1475921720988666>.
- [61] C. Fan, Y. Sun, Y. Zhao, M. Song, J. Wang, Deep learning-based feature engineering methods for improved building energy prediction, *Appl. Energy* 240 (2019) 35–45.
- [62] M. Fey, J.E. Lenssen, Fast graph representation learning with PyTorch Geometric, *arXiv Preprint arXiv:1903.02428* (2019).
- [63] P. Velickovic, G. Cucurull, A. Casanova, A. Romero, P. Lio, Y. Bengio, Graph attention networks, *Stat* 1050 (2017) 10–48550.
- [64] W. Hamilton, Z. Ying, J. Leskovec, Inductive representation learning on large graphs, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [65] X. Wang, H. Ji, C. Shi, B. Wang, Y. Ye, P. Cui, P.S. Yu, Heterogeneous graph attention network, in: 2019: pp. 2022–2032.
- [66] E. Rossi, B. Chamberlain, F. Frasca, D. Eynard, F. Monti, M. Bronstein, Temporal graph networks for deep learning on dynamic graphs, *arXiv Preprint arXiv:2006.10637* (2020).
- [67] S. Yan, Y. Xiong, D. Lin, Spatial temporal graph convolutional networks for skeleton-based action recognition, in: 2018.
- [68] D. Moher, A. Liberati, J. Tetzlaff, D.G. Altman, PRISMA Group, Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement, *PLoS Med.* 6 (2009) e1000097, <https://doi.org/10.1371/journal.pmed.1000097>.
- [69] A. Simons, B. Niehaves, B. Niehaves, K. Reimer, RECONSTRUCTING THE GIANT: ON THE IMPORTANCE OF RIGOUR IN DOCUMENTING THE LITERATURE SEARCH PROCESS, in: ECIS 2009 PROCEEDINGS, 2009. <https://aisel.aisnet.org/ecis2009/161>.
- [70] Y. Levy, T. J. Ellis, A Systems Approach to Conduct an Effective Literature Review in Support of Information Systems Research, *Informing Science: The International Journal of an Emerging Transdiscipline* 9 (2006) 181–212. doi: 10.28945/479.
- [71] H. Cooper, Research Synthesis and Meta-Analysis: A Step-by-Step Approach, SAGE Publications, 2015. <https://psycnet.apa.org/record/2009-06723-000>.
- [72] T. Greenhalgh, R. Peacock, Effectiveness and efficiency of search methods in systematic reviews of complex evidence: Audit of primary sources, *BMJ* 331 (2005) 1064–1065, <https://doi.org/10.1136/bmj.38636.593461.68>.
- [73] F. Özcan, C. Lei, A. Quamar, V. Efthymiou, Semantic enrichment of data for AI applications, in: Proceedings of the Fifth Workshop on Data Management for End-To-End Machine Learning, Association for Computing Machinery, New York, NY, USA, 2021: pp. 1–7. doi: 10.1145/3462462.3468881.
- [74] B. Naticchia, A. Giretti, A. Carbonari, V. Massimo, A Real-Time and Non Intrusive Structural Health Monitoring System for Buildings, 2010. doi: 10.22260/ISARC2010/0033.
- [75] V.M. Karbhari, F. Ansari, *Structural Health Monitoring of Civil Infrastructure Systems*, Elsevier, 2009.
- [76] H. Burak Gunay, W. Shen, G. Newsham, Data analytics to improve building performance: A critical review, *Autom. Constr.* 97 (2019) 96–109, <https://doi.org/10.1016/j.autcon.2018.10.020>.

- [77] D.M. Frangopol, Life-cycle performance, management, and optimisation of structural systems under uncertainty: accomplishments and challenges 1, *Struct. Infrastruct. Eng.* 7 (2011) 389–413, <https://doi.org/10.1080/15732471003594427>.
- [78] Y. Wen, L.C.M. Tang, D.C.W. Ho, A BIM-based space-oriented solution for hospital facilities management, *Facilities* 39 (2021) 689–702, <https://doi.org/10.1108/F-10-2019-0105>.
- [79] M. Belsky, R. Sacks, I. Briklakis, Semantic enrichment for building information modeling, *Comput. Aided Civ. Inf. Eng.* 31 (2016) 261–274, <https://doi.org/10.1111/mice.12128>.
- [80] J. Wu, J. Zhang, New automated BIM object classification method to support BIM interoperability, *J. Comput. Civ. Eng.* 33 (2019) 04019033, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000858](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000858).
- [81] Y. Zou, B.H.W. Guo, E. Papadonikolaki, J. Dimyadi, L. Hou, Lessons learned on adopting automated compliance checking in the AEC industry: A global study, *J. Manag. Eng.* 39 (2023) 04023019, <https://doi.org/10.1061/JMENA.MEENG-5051>.
- [82] C. Zhang, F. Wang, Y. Zou, J. Dimyadi, B.H.W. Guo, L. Hou, Automated UAV image-to-BIM registration for building façade inspection using improved generalised Hough transform, *Autom. Constr.* 153 (2023) 104957, <https://doi.org/10.1016/j.autcon.2023.104957>.
- [83] B. Koo, R. Jung, Y. Yu, I. Kim, A geometric deep learning approach for checking element-to-entity mappings in infrastructure building information models, *J. Comput. Des. Eng.* 8 (2021) 239–250, <https://doi.org/10.1093/jcde/qwaa075>.
- [84] T. Bloch, R. Sacks, Comparing machine learning and rule-based inferencing for semantic enrichment of BIM models, *Autom. Constr.* 91 (2018) 256–272, <https://doi.org/10.1016/j.autcon.2018.03.018>.
- [85] T. Bloch, Connecting research on semantic enrichment of BIM - review of approaches, methods and possible applications, *J. Inform. Technol. Constr.* 27 (2022) 416–440, <https://doi.org/10.36680/j.itcon.2022.020>.
- [86] F.M. Dinis, J. Poças Martins, A.S. Guimarães, B. Rangel, BIM and semantic enrichment methods and applications: A review of recent developments, *Arch. Comput. Meth. Eng.* 29 (2022) 879–895, <https://doi.org/10.1007/s11831-021-09595-6>.
- [87] Q. Lu, X. Xie, J. Heaton, A.K. Parlikad, J. Schooling, From BIM towards digital twin: strategy and future development for smart asset management, Service Oriented, Holonic and Multi-Agent Manufacturing Systems for Industry of the Future: Proceedings of SOHOMA 2019 9 (2020) 392–404.
- [88] A. Ziaee, G. Suter, SAGC-A68: a space access graph dataset for the classification of spaces and space elements in apartment buildings, in: 30 Th International Workshop on Intelligent Computing in Engineering (EG-ICE), 2023. doi: 10.48550/arXiv.2307.04515.
- [89] Z. Zheng, F. Petzold, Neural-guided room layout generation with bubble diagram constraints, *Autom. Constr.* 154 (2023) 104962, <https://doi.org/10.1016/j.autcon.2023.104962>.
- [90] H. Park, H. Suh, J. Kim, S. Choo, Floor plan recommendation system using graph neural network with spatial relationship dataset, *J. Build. Eng.* 71 (2023) 106378, <https://doi.org/10.1016/j.jobe.2023.106378>.
- [91] L. Yang, W. Huang, Representation and assessment of spatial design using a hierarchical graph neural network: Classification of shopping center types, *Autom. Constr.* 147 (2023) 104727, <https://doi.org/10.1016/j.autcon.2022.104727>.
- [92] L. Jiang, J. Shi, C. Wang, Multi-ontology fusion and rule development to facilitate automated code compliance checking using BIM and rule-based reasoning, *Adv. Eng. Inf.* 51 (2022) 101449, <https://doi.org/10.1016/j.aei.2021.101449>.
- [93] H.Y. Quek, M. Hofmeister, S.D. Rihm, J. Yan, J. Lai, G. Brownbridge, M. Hillman, S. Mosbach, W. Ang, Y.-K. Tsai, D.N. Tran, S. Kang, W. Tan, M. Kraft, BIM-GIS Integration: Knowledge graphs in a world of data silos, (n.d.).
- [94] L. Rauch, T. Braml, Semantic point cloud segmentation with deep-learning-based approaches for the construction industry: A survey, *Appl. Sci.* 13 (2023) 9146, <https://doi.org/10.3390/app13169146>.
- [95] A. Baik, From point cloud to jeddah heritage BIM nasif historical house—case study, *Digital Applications in Archaeology and Cultural Heritage* 4 (2017) 1–18.
- [96] Y. Geng, Z. Wang, L. Jia, Y. Qin, Y. Chai, K. Liu, L. Tong, 3DGraphSeg: A unified graph representation-based point cloud segmentation framework for full-range highspeed railway environments, *IEEE Trans. Ind. Inf.* (2023) 1–13, <https://doi.org/10.1109/TII.2023.3246492>.
- [97] X. Yang, E. del Rey Castillo, Y. Zou, L. Wotherspoon, Y. Tan, Automated semantic segmentation of bridge components from large-scale point clouds using a weighted superpoint graph, *Automation in Construction* 142 (2022) 104519. doi: 10.1016/j.autcon.2022.104519.
- [98] J.S. Lee, J. Park, Y.-M. Ryu, Semantic segmentation of bridge components based on hierarchical point cloud model, *Autom. Constr.* 130 (2021) 103847, <https://doi.org/10.1016/j.autcon.2021.103847>.
- [99] Z. Xing, S. Zhao, W. Guo, F. Meng, X. Guo, S. Wang, H. He, Coal resources under carbon peak: Segmentation of massive laser point clouds for coal mining in underground dusty environments using integrated graph deep learning model, *Energy* 285 (2023) 128771, <https://doi.org/10.1016/j.energy.2023.128771>.
- [100] T.-Y. Chuang, C.-C. Sung, Learning-guided point cloud vectorization for building component modeling, *Autom. Constr.* 132 (2021) 103978, <https://doi.org/10.1016/j.autcon.2021.103978>.
- [101] R. Zhai, J. Zou, Y. He, L. Meng, BIM-driven data augmentation method for semantic segmentation in superpoint-based deep learning network, *Autom. Constr.* 140 (2022) 104373, <https://doi.org/10.1016/j.autcon.2022.104373>.
- [102] F.C. Collins, M. Ringsquandl, A. Braun, D.M. Hall, A. Borrmann, Shape encoding for semantic healing of design models and knowledge transfer to scan-to-BIM, *Proceedings of the Institution of Civil Engineers - Smart Infrastructure and Construction* 175 (2022) 160–180. doi: 10.1680/jsmic.21.00032.
- [103] F.C. Collins, A. Braun, A. Borrmann, Finding geometric and topological similarities in building elements for large-scale pose updates in Scan-vs-BIM, in: *Proceedings of the International Conference on Computing in Civil and Building Engineering (ICCCBE)*, 2022. <https://mediatum.ub.tum.de/1703136> (accessed July 25, 2023).
- [104] H. Feng, L. Ma, Y. Yu, Y. Chen, J. Li, SCL-GCN: Stratified contrastive learning graph convolution network for pavement crack detection from mobile LiDAR point clouds, *Int. J. Appl. Earth Obs. Geoinf.* 118 (2023) 103248, <https://doi.org/10.1016/j.jag.2023.103248>.
- [105] L. Ma, J. Li, SD-GCN: Saliency-based dilated graph convolution network for pavement crack extraction from 3D point clouds, *Int. J. Appl. Earth Obs. Geoinf.* 111 (2022) 102836, <https://doi.org/10.1016/j.jag.2022.102836>.
- [106] M.M. Akhlaghi, S. Bose, M.E. Mohammadi, B. Moaveni, A. Stavridis, R.L. Wood, Post-earthquake damage identification of an RC school building in Nepal using ambient vibration and point cloud data, *Eng. Struct.* 227 (2021) 111413.
- [107] S. Tutas, A. Braun, A. Borrmann, U. Stilla, Validation of BIM components by photogrammetric point clouds for construction site monitoring, *ISPRS Ann. Photogram. Remote Sens. Spatial Inform. Sci.* 2 (2015) 231–237.
- [108] M.S. Osman, A.M. Abu-Mahfouz, P.R. Page, A survey on data imputation techniques: Water distribution system as a use case, *IEEE Access* 6 (2018) 63279–63291, <https://doi.org/10.1109/ACCESS.2018.2877269>.
- [109] E. Acuña, C. Rodríguez, The Treatment of Missing Values and its Effect on Classifier Accuracy, in: D. Banks, F.R. McMorris, P. Arabie, W. Gaul (Eds.), *Classification, Clustering, and Data Mining Applications*, Springer, Berlin, Heidelberg, 2004: pp. 639–647. doi: 10.1007/978-3-642-17103-1_60.
- [110] J. Ma, J.C.P. Cheng, F. Jiang, W. Chen, M. Wang, C. Zhai, A bi-directional missing data imputation scheme based on LSTM and transfer learning for building energy data, *Energy Build.* 216 (2020) 109941, <https://doi.org/10.1016/j.enbuild.2020.109941>.
- [111] I. Spinelli, S. Scardapane, A. Uncini, Missing data imputation with adversarially-trained graph convolutional networks, *Neural Netw.* 129 (2020) 249–260, <https://doi.org/10.1016/j.neunet.2020.06.005>.
- [112] J. Niu, S. Li, Z. Li, Restoration of missing structural health monitoring data using spatiotemporal graph attention networks, *Struct. Health Monit.* 21 (2022) 2408–2419, <https://doi.org/10.1177/14759217211056832>.
- [113] S. Bloemheuvel, J. van den Hoogen, M. Atzmueller, A computational framework for modeling complex sensor network data using graph signal processing and graph neural networks in structural health monitoring, *Appl. Network Sci.* 6 (2021) 1–24, <https://doi.org/10.1007/s41109-021-00438-8>.
- [114] L. Xing, L. Sela, Graph neural networks for state estimation in water distribution systems: Application of supervised and semisupervised learning, *J. Water Resour. Plan. Manag.* 148 (2022) 04022018, [https://doi.org/10.1061/\(ASCE\)WR.1943-5452.0001550](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001550).
- [115] Y. Wu, X. Wang, S. Liu, X. Yu, X. Wu, A weighting strategy to improve water demand forecasting performance based on spatial correlation between multiple sensors, *Sustain. Cities Soc.* 93 (2023) 104545, <https://doi.org/10.1016/j.scs.2023.104545>.
- [116] Y. Zhao, S. Wang, F. Xiao, A statistical fault detection and diagnosis method for centrifugal chillers based on exponentially-weighted moving average control charts and support vector regression, *Appl. Therm. Eng.* 51 (2013) 560–572, <https://doi.org/10.1016/j.applthermaleng.2012.09.030>.
- [117] C. Miller, Z. Nagy, A. Schlueter, A review of unsupervised statistical learning and visual analytics techniques applied to performance analysis of non-residential buildings, *Renew. Sustain. Energy Rev.* 81 (2018) 1365–1377, <https://doi.org/10.1016/j.rser.2017.05.124>.
- [118] Y. Himeur, K. Ghanem, A. Alsalemi, F. Bensaali, A. Amira, Artificial intelligence based anomaly detection of energy consumption in buildings: A review, current trends and new perspectives, *Appl. Energy* 287 (2021) 116601, <https://doi.org/10.1016/j.apenergy.2021.116601>.
- [119] Y. Gu, G. Li, J. Gu, J.J. Jung, Graph embedding-based Anomaly localization for HVAC system, *J. Build. Eng.* 77 (2023) 107511, <https://doi.org/10.1016/j.jobe.2023.107511>.
- [120] X. Shu, T. Bao, Y. Zhou, R. Xu, Y. Li, K. Zhang, Unsupervised dam anomaly detection with spatial-temporal variational autoencoder, *Struct. Health Monit.* 22 (2023) 39–55, <https://doi.org/10.1177/14759217211073301>.
- [121] V.-H. Dang, K. Le-Nguyen, T.-T. Nguyen, Semi-supervised vibration-based structural health monitoring via deep graph learning and contrastive learning, *Structures* 51 (2023) 158–170, <https://doi.org/10.1016/j.istruc.2023.03.011>.
- [122] W. Kim, S. Katipamula, A review of fault detection and diagnostics methods for building systems, *Sci. Technol. Built Environ.* 24 (2018) 3–21, <https://doi.org/10.1080/23744731.2017.1318008>.
- [123] M.R. Sanzana, T. Maul, J.Y. Wong, M.O.M. Abdulrazic, C.-C. Yip, Application of deep learning in facility management and maintenance for heating, ventilation, and air conditioning, *Autom. Constr.* 141 (2022) 104445, <https://doi.org/10.1016/j.autcon.2022.104445>.
- [124] C. Fan, Y. Lin, M.S. Piscitelli, R. Chiosa, H. Wang, A. Capozzoli, Y. Ma, Leveraging graph convolutional networks for semi-supervised fault diagnosis of HVAC systems in data-scarce contexts, *Build. Simul.* (2023), <https://doi.org/10.1007/s12273-023-1041-1>.
- [125] J. Zhang, Y. Cheng, X. He, Fault diagnosis of energy networks: A graph embedding learning approach, *IEEE Trans. Instrum. Meas.* 71 (2022), <https://doi.org/10.1109/TIM.2022.3216669>.
- [126] J.B. Haurum, M. Madadi, S. Escalera, T.B. Moeslund, Multi-Task Classification of Sewer Pipe Defects and Properties using a Cross-Task Graph Neural Network

- Decoder, in: 2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), IEEE, Waikoloa, HI, USA, 2022: pp. 1441–1452. doi: 10.1109/WACV51458.2022.00151.
- [127] P. Zhan, X. Qin, Q. Zhang, Y. Sun, A novel structural damage detection method via multisensor spatial-temporal graph-based features and deep graph convolutional network, *IEEE Trans. Instrum. Measur.* 72 (2023) 1–14, <https://doi.org/10.1109/TIM.2023.3238048>.
- [128] X. Huang, X. Zhou, R. Duan, Concrete Crack Detection Based on Hybrid Residual Network and Graph Convolutional Network, in: Qin Y., Jia L., Liang J., Liu Z., Diao L., An M. (Eds.), *Lecture Notes in Electrical Engineering*, Springer Science and Business Media Deutschland GmbH, 2022: pp. 74–81. doi: 10.1007/978-981-16-9913-9_9.
- [129] X. Yan, X. Song, Large-scale civil engineering structure deformation monitoring research based on image recognition, *Traitement Du Signal* 40 (2023) 501–509, <https://doi.org/10.18280/ts.400209>.
- [130] C. Deb, F. Zhang, J. Yang, S.E. Lee, K.W. Shah, A review on time series forecasting techniques for building energy consumption, *Renew. Sustain. Energy Rev.* 74 (2017) 902–924, <https://doi.org/10.1016/j.rser.2017.02.085>.
- [131] M. Bourdeau, X. Qiang Zhai, E. Nefzaoui, X. Guo, P. Chatellier, Modeling and forecasting building energy consumption: A review of data-driven techniques, *Sustainable Cities and Society* 48 (2019) 101533. doi: 10.1016/j.scs.2019.101533.
- [132] X.J. Luo, L.O. Oyedele, Forecasting building energy consumption: Adaptive long-short term memory neural networks driven by genetic algorithm, *Adv. Eng. Inf.* 50 (2021) 101357, <https://doi.org/10.1016/j.aei.2021.101357>.
- [133] D. Enescu, A review of thermal comfort models and indicators for indoor environments, *Renew. Sustain. Energy Rev.* 79 (2017) 1353–1379, <https://doi.org/10.1016/j.rser.2017.05.175>.
- [134] J. Cao, H. Zhang, A. Savov, D. Hall, B. Dillenburger, Energy-aware design: predicting building performance from layout graphs, in: University of Turin, 2022: pp. 0–0. doi: 10.35490/EC3.2022.210.
- [135] Y. Hu, X. Cheng, S. Wang, J. Chen, T. Zhao, E. Dai, Times series forecasting for urban building energy consumption based on graph convolutional network, *Appl. Energy* 307 (2022) 118231, <https://doi.org/10.1016/j.apenergy.2021.118231>.
- [136] X. Wang, X. Wang, X. Yin, K. Li, L. Wang, R. Wang, R. Song, Distributed LSTM-GCN based spatial-temporal indoor temperature prediction in multi-zone buildings, *IEEE Trans. Ind. Inf.* (2023) 1–10, <https://doi.org/10.1109/TII.2023.3268467>.
- [137] C. Zhou, Z. Xu, J. Wu, K. Liu, X. Guan, Multi-zone indoor temperature prediction based on Graph Attention Network and Gated Recurrent Unit, in: IEEE International Conference on Automation Science and Engineering, IEEE Computer Society, 2021: pp. 1973–1978. doi: 10.1109/CASE49439.2021.9551630.
- [138] J. Zhang, F. Xiao, A. Li, T. Ma, K. Xu, H. Zhang, R. Yan, X. Fang, Y. Li, D. Wang, Graph neural network-based spatio-temporal indoor environment prediction and optimal control for central air-conditioning systems, *Build. Environ.* 242 (2023) 110600, <https://doi.org/10.1016/j.buildenv.2023.110600>.
- [139] Z. Wang, X. Liu, Y. Huang, P. Zhang, Y. Fu, A multivariate time series graph neural network for district heat load forecasting, *Energy* 278 (2023) 127911, <https://doi.org/10.1016/j.energy.2023.127911>.
- [140] W. Zhang, Y. Wu, J.K. Calautit, A review on occupancy prediction through machine learning for enhancing energy efficiency, air quality and thermal comfort in the built environment, *Renew. Sustain. Energy Rev.* 167 (2022) 112704, <https://doi.org/10.1016/j.rser.2022.112704>.
- [141] Z.D. Tekler, A. Chong, Occupancy prediction using deep learning approaches across multiple space types: A minimum sensing strategy, *Build. Environ.* 226 (2022) 109689, <https://doi.org/10.1016/j.buildenv.2022.109689>.
- [142] M.M. Abdelrahman, C. Miller, Targeting occupant feedback using digital twins: Adaptive spatial-temporal thermal preference sampling to optimize personal comfort models, *Build. Environ.* 218 (2022) 109090, <https://doi.org/10.1016/j.buildenv.2022.109090>.
- [143] P. Liu, T. Zhao, J. Luo, B. Lei, M. Frei, F. Biljecki, Towards human-centric digital twins: Leveraging computer vision and graph models to predict outdoor comfort, *Sustain. Cities Soc.* 93 (2023) 104480, <https://doi.org/10.1016/j.scs.2023.104480>.
- [144] Y. Xie, S. Stravaravdis, Generating occupancy profiles for building simulations using a hybrid GNN and LSTM framework, *Energies* 16 (2023) 4638, <https://doi.org/10.3390/en16124638>.
- [145] N. Kayhani, B. McCabe, B. Sankaran, Semantic-aware quality assessment of building elements using graph neural networks, *Autom. Constr.* 155 (2023) 105054, <https://doi.org/10.1016/j.autcon.2023.105054>.
- [146] H. Feng, W. Li, Z. Luo, Y. Chen, S.N. Fatholahi, M. Cheng, C. Wang, J.M. Junior, J. Li, GCN-based pavement crack detection using mobile LIDAR point clouds, *IEEE Trans. Intell. Transp. Syst.* 23 (2022) 11052–11061, <https://doi.org/10.1109/TITS.2021.3099023>.
- [147] H. Kiavarz, M. Jadidi, A. Rajabifard, G. Sohn, ROOM-BASED ENERGY DEMAND CLASSIFICATION OF BIM DATA USING GRAPH SUPERVISED LEARNING, *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences XLVI-4-W4-2021* (2021) 97–100. doi: 10.5194/isprs-archives-XLVI-4-W4-2021-97-2021.
- [148] X. Li, T. Zhao, J. Zhang, T. Chen, Predication control for indoor temperature time-delay using Elman neural network in variable air volume system, *Energ. Build.* 154 (2017) 545–552, <https://doi.org/10.1016/j.enbuild.2017.09.005>.
- [149] Z. Ying, D. Bourgeois, J. You, M. Zitnik, J. Leskovec, Gnnexplainer: Generating explanations for graph neural networks, *Adv. Neural Inf. Process. Syst.* 32 (2019), <https://doi.org/10.48550/arXiv.1903.03894> Focus to learn more.
- [150] H. Cho, E.K. Lee, I.S. Choi, Layer-wise relevance propagation of InteractionNet explains protein-ligand interactions at the atom level, *Sci. Rep.* 10 (2020) 21155.
- [151] G. Suter, Modeling multiple space views for schematic building design using space ontologies and layout transformation operations, *Autom. Constr.* 134 (2022) 104041, <https://doi.org/10.1016/j.autcon.2021.104041>.
- [152] Z. Wang, B. Ouyang, R. Sacks, Graph-based inter-domain consistency maintenance for BIM models, *Autom. Constr.* (2023), <https://doi.org/10.1016/j.autcon.2023.104979>.
- [153] Y. Hu, D. Castro-Lacouture, C.M. Eastman, Holistic clash detection improvement using a component dependent network in BIM projects, *Autom. Constr.* 105 (2019) 102832, <https://doi.org/10.1016/j.autcon.2019.102832>.
- [154] Y. Hu, C. Xia, J. Chen, X. Gao, Clash context representation and change component prediction based on graph convolutional network in MEP disciplines, *Adv. Eng. Inf.* 55 (2023) 101896, <https://doi.org/10.1016/j.aei.2023.101896>.
- [155] V.J.L. Gan, BIM-based graph data model for automatic generative design of modular buildings, *Autom. Constr.* 134 (2022) 104062, <https://doi.org/10.1016/j.autcon.2021.104062>.
- [156] H. Xiong, G. Shen, X. Lan, H. Yuan, X. Kong, HIT-GCN: Spatial-temporal graph convolutional network embedded with heterogeneous information of road network for traffic forecasting, *Electronics* 12 (2023) 1306, <https://doi.org/10.3390/electronics12061306>.
- [157] S. Yang, W. Chen, M.P. Wan, A machine-learning-based event-triggered model predictive control for building energy management, *Build. Environ.* 233 (2023) 110101, <https://doi.org/10.1016/j.buildenv.2023.110101>.
- [158] S. Zhang, Y. Liu, Y. Sun, N. Shah, Graph-less Neural Networks: Teaching Old MLPs New Tricks via Distillation, (2022). doi: 10.48550/arXiv.2110.08727.
- [159] N. Li, Y. Zhao, D. Li, W. Guan, Real-time Prediction of Highway Equipment Faults Based on GCN and GRU Algorithms, in: 2021 IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2021: pp. 3252–3257. doi: 10.1109/SMC52423.2021.9658637.
- [160] M.Z. Naser, An engineer's guide to eXplainable Artificial Intelligence and Interpretable Machine Learning: Navigating causality, forced goodness, and the false perception of inference, *Autom. Constr.* 129 (2021) 103821, <https://doi.org/10.1016/j.autcon.2021.103821>.
- [161] X. Li, F. Zhang, Q. Li, B. Zhou, J. Bao, Exploiting a knowledge hypergraph for modeling multi-nary relations in fault diagnosis reports, *Adv. Eng. Inf.* 57 (2023) 102084, <https://doi.org/10.1016/j.aei.2023.102084>.
- [162] B. Zhou, J. Li, X. Li, B. Hua, J. Bao, Leveraging on causal knowledge for enhancing the root cause analysis of equipment spot inspection failures, *Adv. Eng. Inf.* 54 (2022) 101799, <https://doi.org/10.1016/j.aei.2022.101799>.
- [163] M. Pritoni, D. Paine, G. Fierro, C. Mosiman, M. Poplawski, A. Saha, J. Bender, J. Granderson, Metadata schemas and ontologies for building energy applications: A critical review and use case analysis, *Energies* 14 (2021) 2024.
- [164] L. Xia, Y. Liang, J. Leng, P. Zheng, Maintenance planning recommendation of complex industrial equipment based on knowledge graph and graph neural network, *Reliab. Eng. Syst. Saf.* 232 (2023) 109068, <https://doi.org/10.1016/j.res.2022.109068>.
- [165] C. Baudrit, F. Taillandier, C. Curt, Q.A. Hoang, Z.-M. Sbartai, D. Breyse, Graph based knowledge models for capitalizing, predicting and learning: A proof of concept applied to the dam systems, *Adv. Eng. Inf.* 52 (2022) 101551, <https://doi.org/10.1016/j.aei.2022.101551>.
- [166] C. He, M. Liu, S.M. Hsiang, N. Pierce, Synthesizing ontology and graph neural network to unveil the implicit rules for US bridge preservation decisions, *J. Manag. Eng.* 40 (2024) 04024007, <https://doi.org/10.1061/JMENEA.MEENG-5803>.
- [167] Y. Li, L. Liu, G. Wang, Y. Du, P. Chen, EGNN: Constructing explainable graph neural networks via knowledge distillation, *Knowl.-Based Syst.* 241 (2022) 108345, <https://doi.org/10.1016/j.knosys.2022.108345>.
- [168] A. Bellagarda, S. Cesari, A. Aliberti, F. Ugliotti, L. Bottaccioli, E. Macii, E. Patti, Effectiveness of neural networks and transfer learning for indoor air-temperature forecasting, *Autom. Constr.* 140 (2022) 104314, <https://doi.org/10.1016/j.autcon.2022.104314>.
- [169] K. Duan, C.W.K. Suen, Z. Zou, Robot morphology evolution for automated HVAC system inspections using graph heuristic search and reinforcement learning, *Autom. Constr.* 153 (2023) 104956, <https://doi.org/10.1016/j.autcon.2023.104956>.
- [170] G. Morcous, Z. Lounis, Maintenance optimization of infrastructure networks using genetic algorithms, *Autom. Constr.* 14 (2005) 129–142.
- [171] H.H. Hosamo, H.K. Nielsen, D. Kraniotis, P.R. Svennevig, K. Svidt, Improving building occupant comfort through a digital twin approach: A Bayesian network model and predictive maintenance method, *Energ. Build.* 288 (2023) 112992, <https://doi.org/10.1016/j.enbuild.2023.112992>.
- [172] C.-Y. Liu, J.-S. Chou, Bayesian-optimized deep learning model to segment deterioration patterns underneath bridge decks photographed by unmanned

- aerial vehicle, *Autom. Constr.* 146 (2023) 104666, <https://doi.org/10.1016/j.autcon.2022.104666>.
- [173] H. Morgenstern, M. Raupach, Predictive BIM with integrated bayesian inference of deterioration models as a four-dimensional decision support tool, *Civil Eng.* 4 (2023) 185–203, <https://doi.org/10.3390/civileng4010012>.
- [174] S. Yun, M. Jeong, R. Kim, J. Kang, H.J. Kim, Graph transformer networks, *Adv. Neural Inf. Proces. Syst.* 32 (2019).
- [175] Y. Wu, H.-N. Dai, H. Tang, Graph neural networks for anomaly detection in industrial internet of things, *IEEE Internet Things J.* 9 (2021) 9214–9231.
- [176] T. Mortlock, D. Muthirayan, S.-Y. Yu, P.P. Khargonekar, M. Abdullah Al Faruque, Graph Learning for Cognitive Digital Twins in Manufacturing Systems, *IEEE Transactions on Emerging Topics in Computing* 10 (2022) 34–45. doi: 10.1109/TETC.2021.3132251.